MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, D/A, KEY CODE MKR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establi nes the requirements and test procedures for the acquisition of an electronic D/A converter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.37 failures per million operating hours for class I or 2.15 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MKR.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/333-1 for class I and M28787/333-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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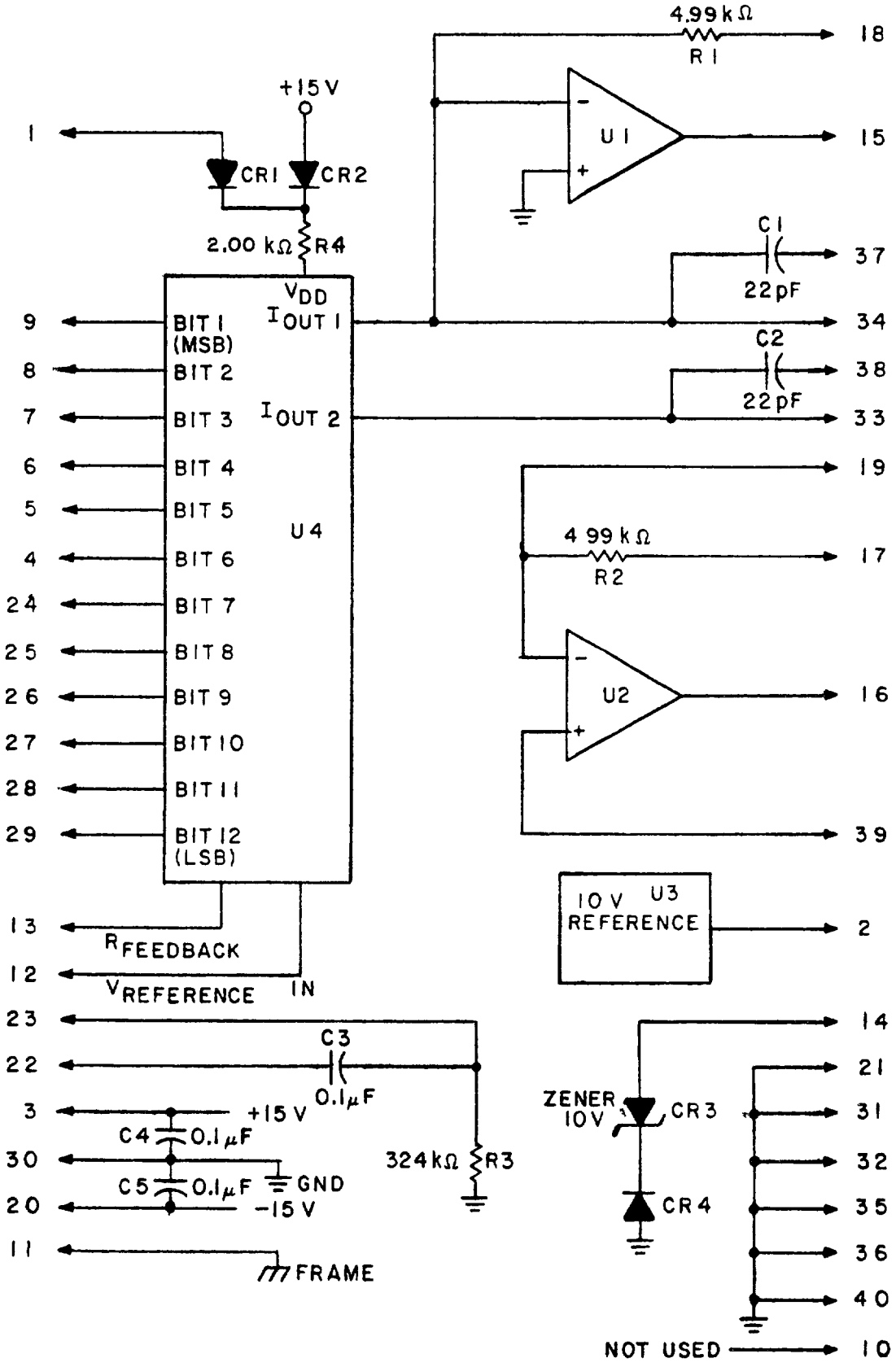


FIGURE 1. MKR function diagram, D/A converter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B12	Input clamp diode voltage, negative	V	X	-0.5	-1.0	-0.5	-1.1	-0.3	-1.0	-0.5	-1.2	-0.3	-1.0	V
C1, C2	+15 Power supply current	VI	S	1	23	1	24	1	23	1	27	1	23	mA
C3, C4	-15 Power supply current	VI	S	1	18	1	19	1	18	1	22	1	18	mA
C5	+5 Power supply current, positive supply connected	VI	S	-1	1	-1	1	-1	1	-1	1	-1	1	µA
C6	+5 Power supply current, positive supply not connected	VI	S	1	1500	1	1500	1	1500	1	1500	1	1500	µA
D1, D2	Reference voltage, voltage regulation	VII	X	9.969	10.031	9.954	10.046	9.954	10.046	9.939	10.061	9.939	10.061	V
D3	Reference voltage, load regulation	VII	X	9.966	10.031	9 951	10.046	9.951	10.046	9.936	10.061	9.936	10.061	V
E	Offset voltage, op amp	VIII	X	-5	5	-7	7	-7	7	-7	7	-7	7	mV
F1, F2	Capacitance, power supply	IX	X	0.075	0.125	0.075	0.125	0.075	0.125	0.075	0.125	0.075	0.125	µF
F3	Capacitance, C3	IX	X	0.089	0.111	0.075	0 125	0.075	0.125	0.075	0.125	0.075	0.125	µF
F4, F5	Capacitance, C1 and C2	IX	X	19	30	16	35	16	35	16	35	16	35	pF
G1	Resistance, R3	X	X	317	331	316	332	316	332	315	333	315	333	kΩ
G2-G8	Continuity	X	X	0	1	0	1	0	1	0	1	0	1	Ω
H1	Diode network, zener voltage	XI	X	-9.5	-11.3	-9.4	-11.3	-9.5	-11.6	-9.1	-11.3	-9.5	-11.8	V
H2	Diode network, diode reverse bias	XI	X	19.8	20.2	19.8	20.2	19.8	20.2	19.8	20.2	19.8	20.2	V

See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1	Bipolar accuracy, positive full scale	XII	X	9.72	10.27	9.63	10.37	9.63	10.37	9.57	10.43	9.57	10.43	V
J2	Bipolar accuracy, zero scale	XII	X	-0.14	0.14	-0.18	0.18	-0.18	0.18	-0.21	0.21	-0.21	0.21	V
J3	Bipolar accuracy, negative full scale	XII	X	-9.93	-10.07	-9.90	-10.10	-9.90	-10.10	-9.89	-10.11	-9.89	-10.11	V
K1	DAC unipolar offset voltage	XIII	X	-5	5	-7	7	-7	7	-7	7	-7	7	mV
K2	DAC unipolar full scale error	XIII	X	-9.93	-10.07	-9.90	-10.10	-9.90	-10.10	-9.89	-10.11	-9.89	-10.11	V
L1-L38	DAC linearity check	XIV	X	-0.8	0.8	-0.9	0.9	-0.9	0.9	-1	1	-1	1	LSB
M1-M15	DAC differential linearity error check	XV	X	-1	1	-1.0	1.2	-1.0	1.2	-2	2	-1.0	1.3	LSB
N1	Gain frequency response check (10 kHz)	XVI	X	35.6	36.4	35.6	36.4	35.4	36.4	35.6	36.4	35.2	36.4	dB
N2	Gain frequency response check (30 kHz)	XVI	X	29.2	36.4	29.2	36.4	29.0	36.4	29.2	36.4	28.8	36.4	dB
N3	Gain frequency response check (100 kHz)	XVI	X	24.4	36.4	24.4	36.4	24.2	36.4	24.4	36.4	24.0	36.4	dB
P	Distortion	XVII	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	%
Q1, Q2	V_{REF} input impedance check	XVIII	S	5	20	5	20	5	20	5	20	5	20	kΩ
R1, R2	Settling time	XIX	S	0.0	5.5	0.0	5.5	0.0	5.5	0.0	5.5	0.0	5.5	µs
S	Power supply rejection ratio check	XX	S	-38	-∞	-38	-∞	-38	-∞	-38	-∞	-38	-∞	dB
T	AC feedthrough check	XXI	S	-60	-∞	-60	-∞	-60	-∞	-60	-∞	-60	-∞	dB
U1	Noise, op amp (20 Hz)	XXII	S	-110	-∞	-110	-∞	-110	-∞	-110	-∞	-110	-∞	dBV
U2	Noise, D/A converter (20 Hz)	XXII	S	-90	-∞	-90	-∞	-90	-∞	-90	-∞	-90	-∞	dBV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U3	Noise, op amp (100 Hz)	XXII	S	-120	-∞	-120	-∞	-120	-∞	-120	-∞	-120	-∞	dBV
U4	Noise, D/A converter (100 Hz)	XXII	S	-100	-∞	-100	-∞	-100	-∞	-100	-∞	-100	-∞	dBV
U5,U7,U9	Noise, op amp (1 kHz, 10 kHz, 40 kHz)	XXII	S	-130	-∞	-130	-∞	-130	-∞	-130	-∞	-130	-∞	dBV
U6,U8,U10	Noise, D/A converter (1 kHz, 10 kHz, 40 kHz)	XXII	S	-110	-∞	-110	-∞	-110	-∞	-110	-∞	-110	-∞	dBV
V1-V12	Digital input current, high level	XXIII	X	-1	1	---	---	-1	1	---	---	-1	1	µA
W1-W12	Digital input current, low level	XXIV	X	-1	1	---	---	-1	1	---	---	-1	1	µA
X1-X12	Digital input capacitance	XXV	Q	2	15	2	15	2	15	2	15	2	15	pF
Y	Output latch	Fig 4	Q	-0.5	0.5	---	---	-0.5	0.5	---	---	-0.5	0.5	V
Z1-Z6	Power supply sequence	XXVI	X	-9.8	-10.2	-9.8	-10.2	-9.8	-10.2	-9.8	-10.2	-9.8	-10.2	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X and Z shall be performed in any order with the following exceptions. If test conditions A, B, or Z are part of a sequence of tests, the order of testing shall be as follows: A, B, Z where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X and Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test and prior to disconnecting the power or reducing the temperature, electrical test Y shall be performed in accordance with figure 4. While still at the maximum class temperature, electrical tests C through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits from table I.

	Class I	Class II	Units
Maximum power dissipation	.71	.81	W
Input voltage:			
Low level	0.8 max	0.8 max	V
High level	2.4 min	2.4 min	V
Input current:			
Low level (V_{IL} = 0.4 V)	-1 min, 1 max	-1 min, 1 max	µA
High level (V_{IH} = 2.4 V)	-1 min, 1 max	-1 min, 1 max	µA
Unipolar accuracy gain error	±1 (< ±1/2 LSB in 5 bits)	±1.1 (< ±1/2 LSB in 5 bits)	%
Offset voltage	-7 min, 7 max	-7 min, 7 max	mV

6.5 Module function. The module contains one 12 bit multiplying digital-to-analog converter (DAC), one 10 volt reference, two operational amplifiers, a high pass filter, and a zener clamping network.

6.5.1 Unipolar input to output relationship. Table II shows the digital input to analog output relationship. Module pin connections for this operation are as follows:

Digital inputs: 4-9 and 24-29.
Connect: 13 to 15 (feedback).
2 to 12 (if 10 V reference is required).
16 to 17 (to keep U2 from oscillating).
Ground to: 33 and 39.
Analog output: 15.

TABLE II. Unipolar input to output relationship.

Digital input			Analog output
1111	1111	1111	$-V_{REF}(1-2^{-12})$
1000	0000	0001	$-V_{REF}(1/2 + 2^{-12})$
1000	0000	0000	$-V_{REF}/2$
0111	1111	1111	$-V_{REF}(1/2 - 2^{-12})$
0000	0000	0001	$-V_{REF}(2^{-12})$
0000	0000	0000	0

1 LSB = 2^{-12} V_{REF}

6.5.2 Bipolar input to output relationship. Table III shows the digital input to analog output relationship. Module pin connections for this operation are as follows:

Digital inputs: 4-9 and 24-29.
Connect: 13 to 15 (feedback).
19 to 33.
16 to 17 to 18.
2 to 12 (if 10 V reference is required).
Ground to: 39.
Analog output: 15.

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TABLE III. Bipolar input to output relationship.

Digital input			Analog output
1111	1111	1111	$-V_{REF}(1-2^{-11})$
1000	0000	0001	$-V_{REF}(2^{-11})$
1000	0000	0000	0
0111	1111	1111	$V_{REF}(2^{-11})$
0000	0000	0001	$V_{REF}(1-2^{-11})$
0000	0000	0000	V_{REF}

1 LSB = 2^{-11} V_{REF}

6.5.3 High pass filter. A single pole, high pass filter can be used to block dc voltages utilizing pin 22 as the input and pin 23 as the output. The -3 dB frequency is at 4.9 Hz.

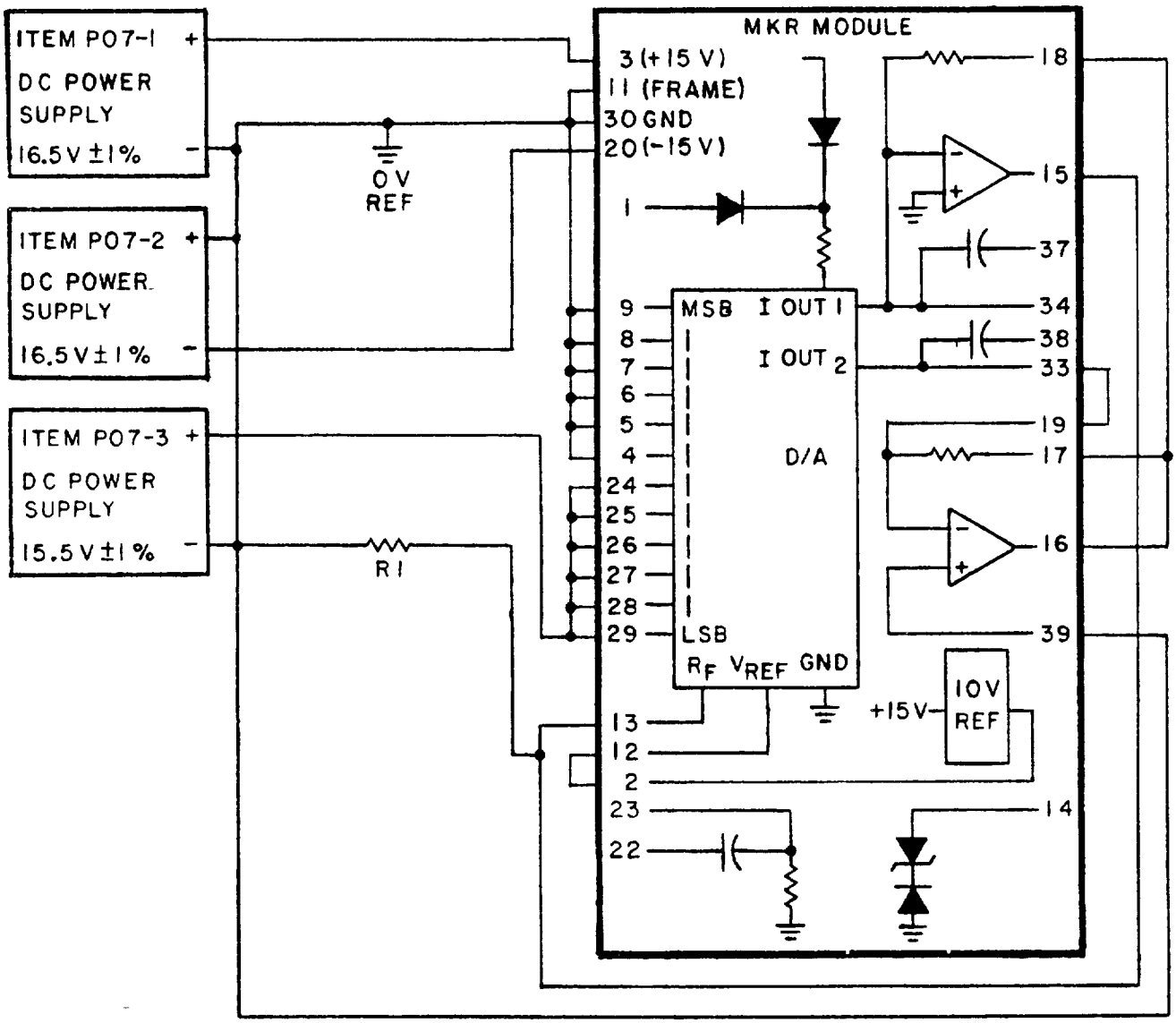
6.5.4 Zener clamping network. The zener clamping network can be used to prevent op amp malfunction when the high pass filter immediately proceeds the op amp U2. Connect pin 14 to pin 23 and the op amp input junction. This will prevent the op amp input from going below approximately -10.3 V.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N305)

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NOTE: R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 2. Operating and component temperature test circuit.

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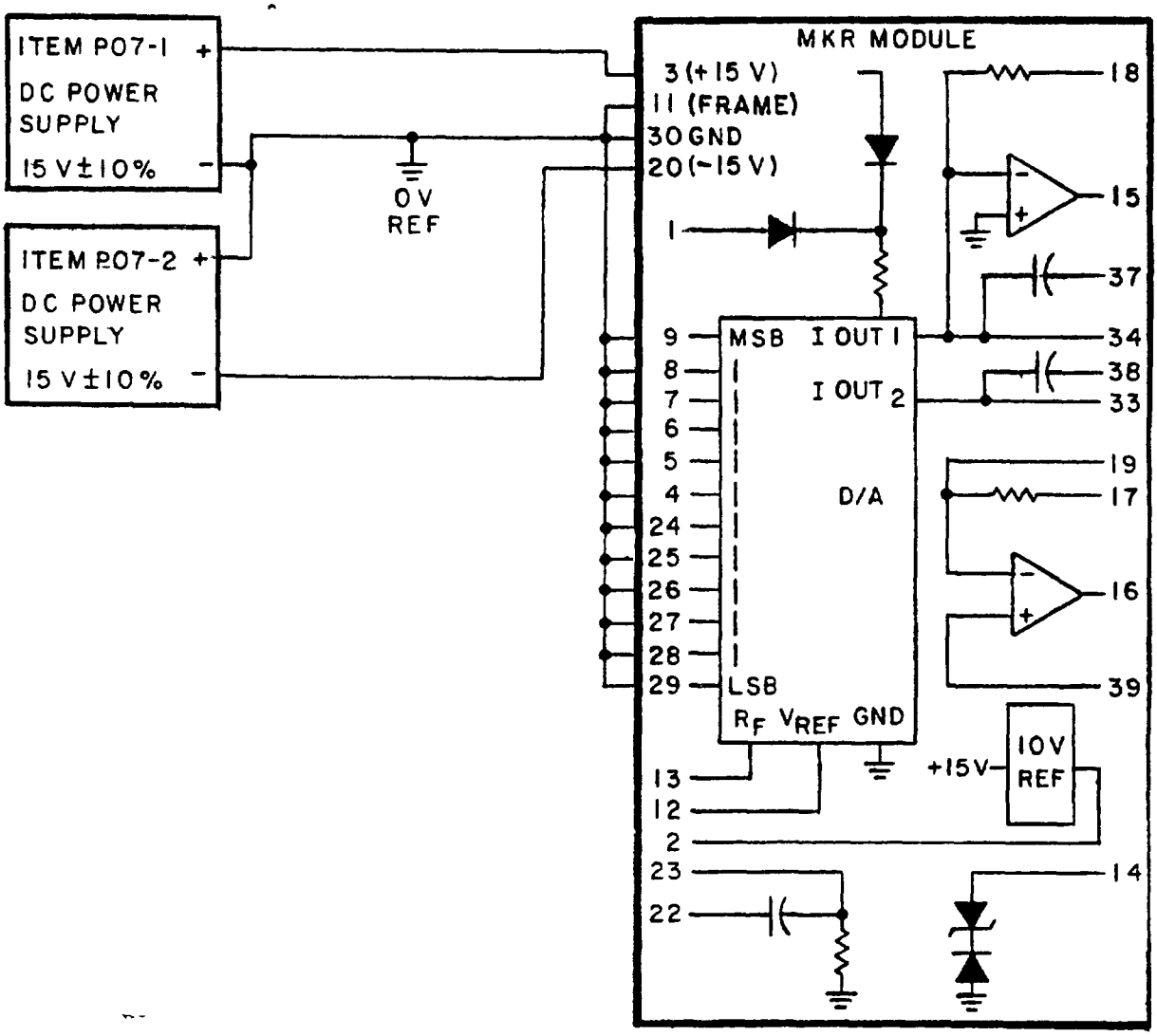
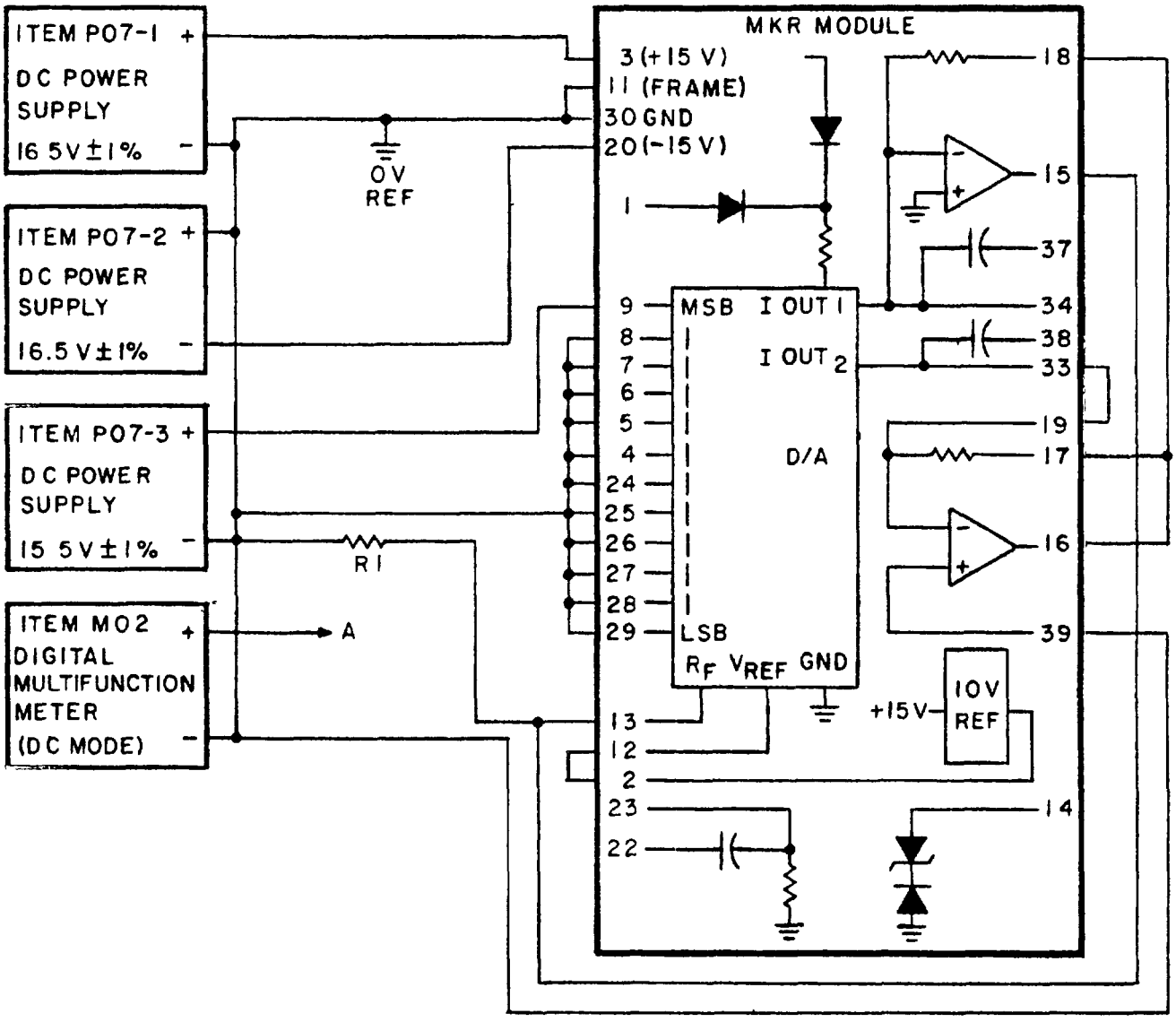


FIGURE 3. Vibration test circuit.

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. Following the life test period, prior to disconnecting the power, connect point A to pin 15. The voltage shall meet the limits specified in table I for test condition Y.
3. Test condition Y verifies that the amplifiers do not latch up.

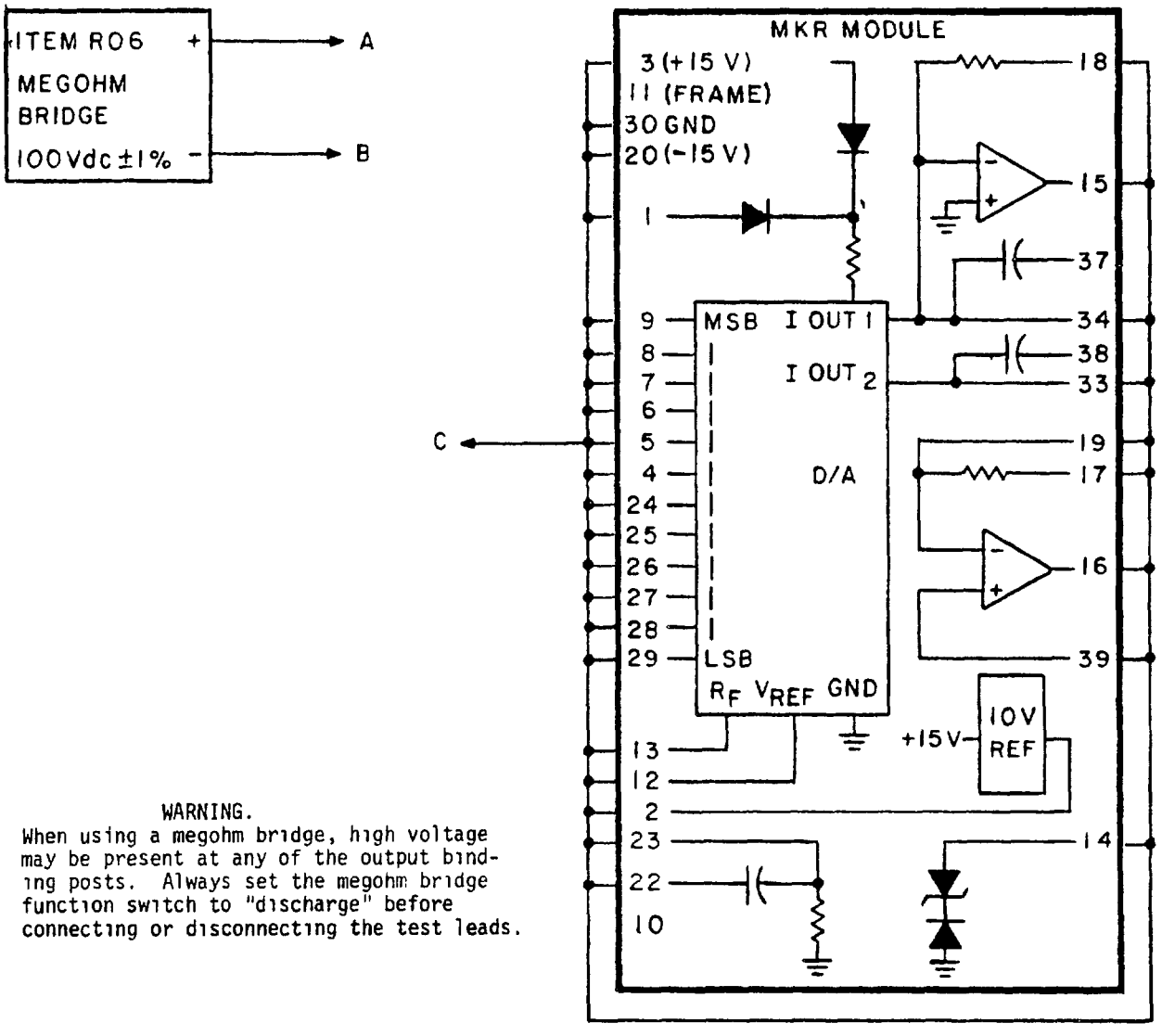
FIGURE 4. Life and output latch test circuit.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C	11	10	∞	MΩ	5
A2	Isolation, used to unused pin	C, 11	10	10	∞	MΩ	5

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WARNING.
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

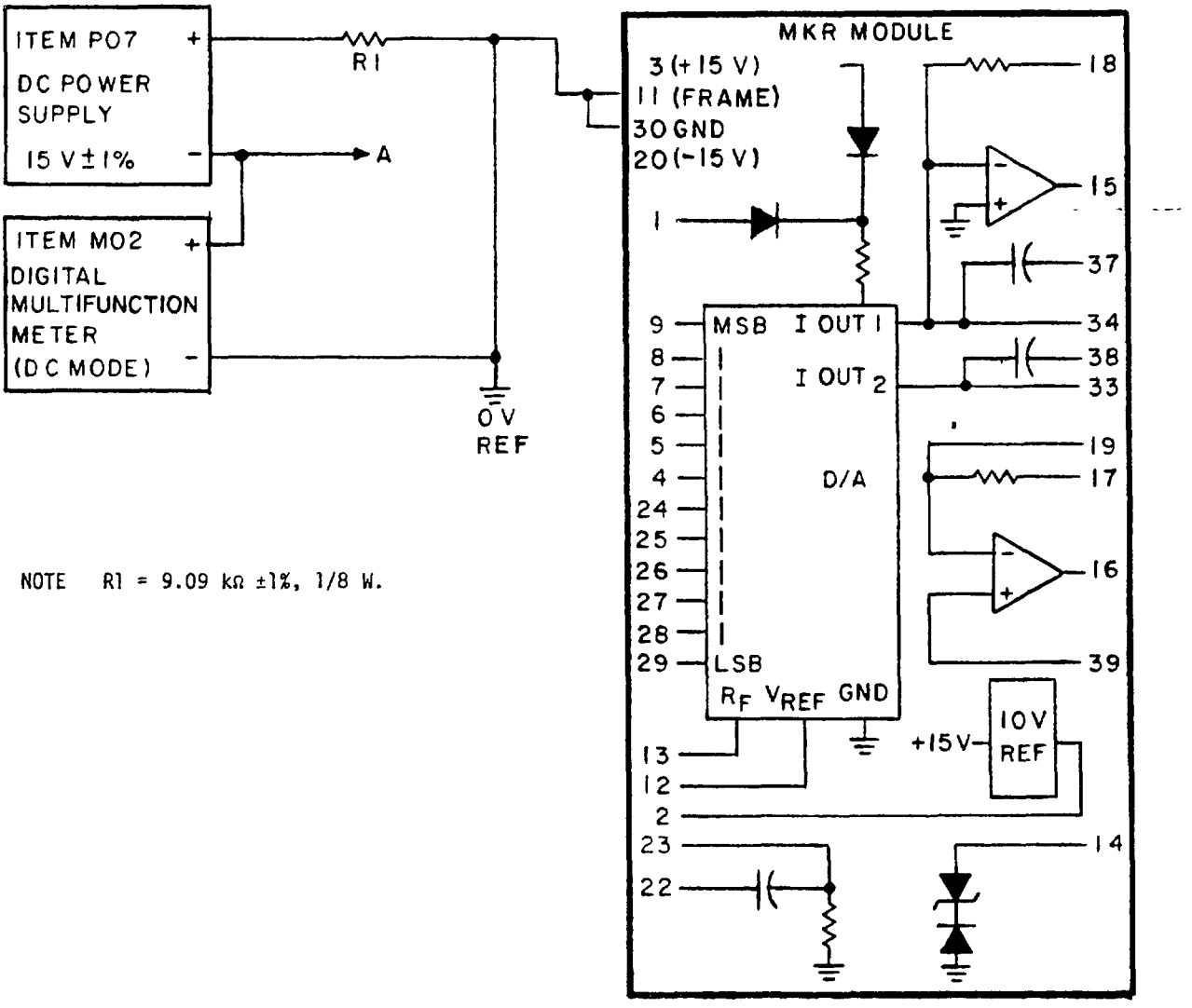
FIGURE 5. Isolation test circuit for table IV.

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TABLE V. Input clamp diode voltage (negative) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
B1	Bit 6 Input clamp diode voltage, negative	4	-0.5	-1.0	V	6
B2	Bit 5	5				
B3	Bit 4	6				
B4	Bit 3	7				
B5	Bit 2	8				
B6	Bit 1	9				
B7	Bit 7	24				
B8	Bit 8	25				
B9	Bit 9	26				
B10	Bit 10	27				
B11	Bit 11	28				
B12	Bit 12	29				

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NOTE R1 = 9.09 kΩ ±1%, 1/8 W.

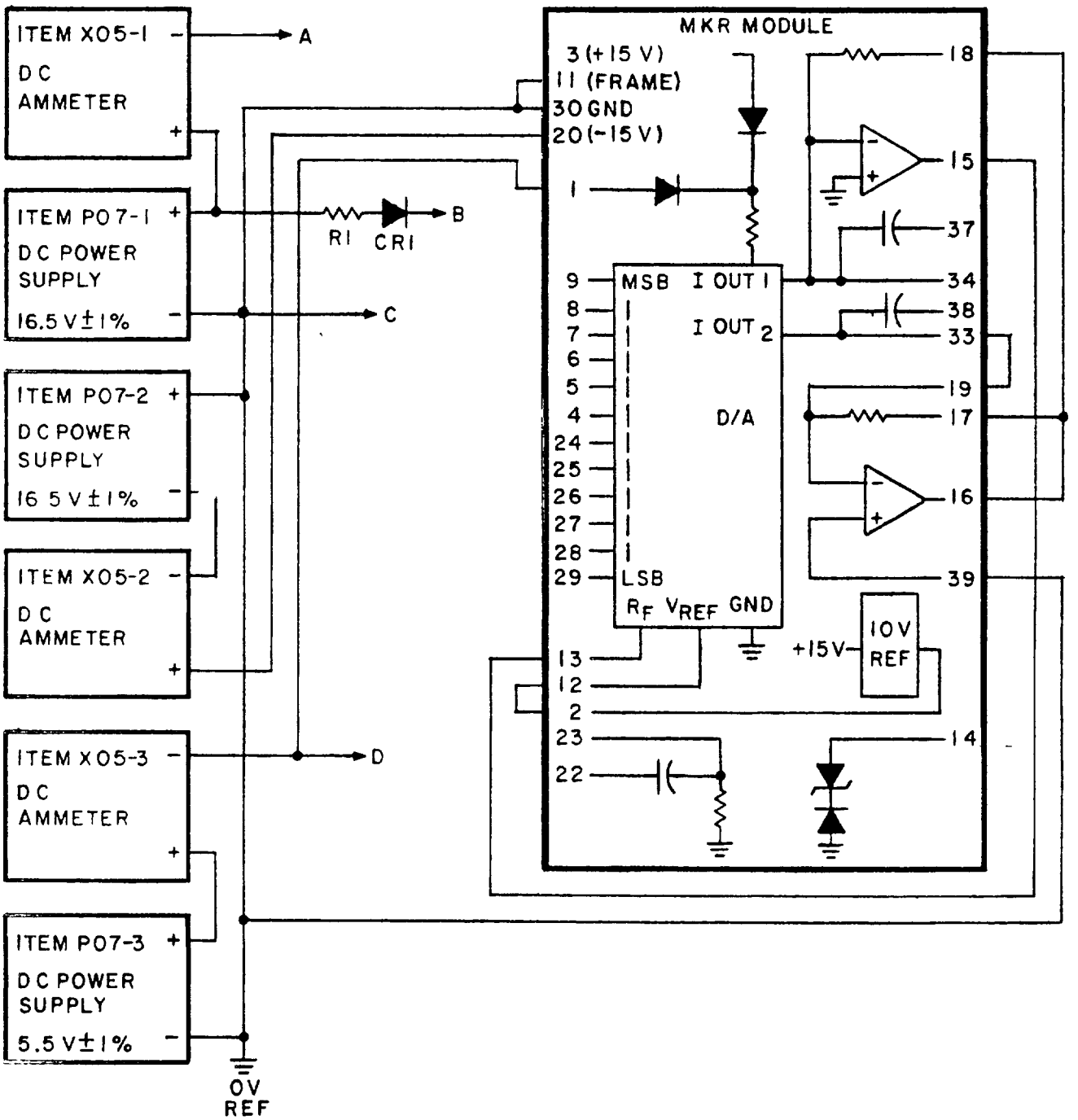
FIGURE 6. Input clamp diode voltage (negative) test circuit for table V.

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TABLE VI. Power supply current test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
C1	+15 V Power supply current, low level	3	Open	4,5,6,7,8,9,24,25,26,27, 28,29	Open	1	23	mA	7
C2	+15 V Power supply current, high level		Open	Open	4,5,6,7, 8,9,24, 25,26, 27,28,29		23		
C3	-15 V Power supply current, low level		Open	4,5,6,7,8,9,24,25,26,27, 28,29	Open		18		
C4	-15 V Power supply current, high level		4,5,6, 7,8,9, 24,25, 26,27, 28,29	Open			18		
C5	+5 Power supply current, (positive supply connected)		Open	4,5,6,7,8,9,24,25,26,27, 28,29		-1	+1	µA	
C6	+5 Power supply current, (positive supply not connected)	Open	Open	4,5,6,7,8,9,24,25,26,27, 28,29		1	1500	µA	

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NOTES:
1. R1 = 3.01 kΩ ±1%, 1/8 W.
2. CR1 = 1N4148 or equivalent (used to keep voltage level at digital input pins approximately the same as D/A converter V_{DD}).
3. For test conditions C1 and C2 read item XO5-1.
   For test conditions C3 and C4 read item XO5-2.
   For test conditions C5 and C6 read item XO5-3.

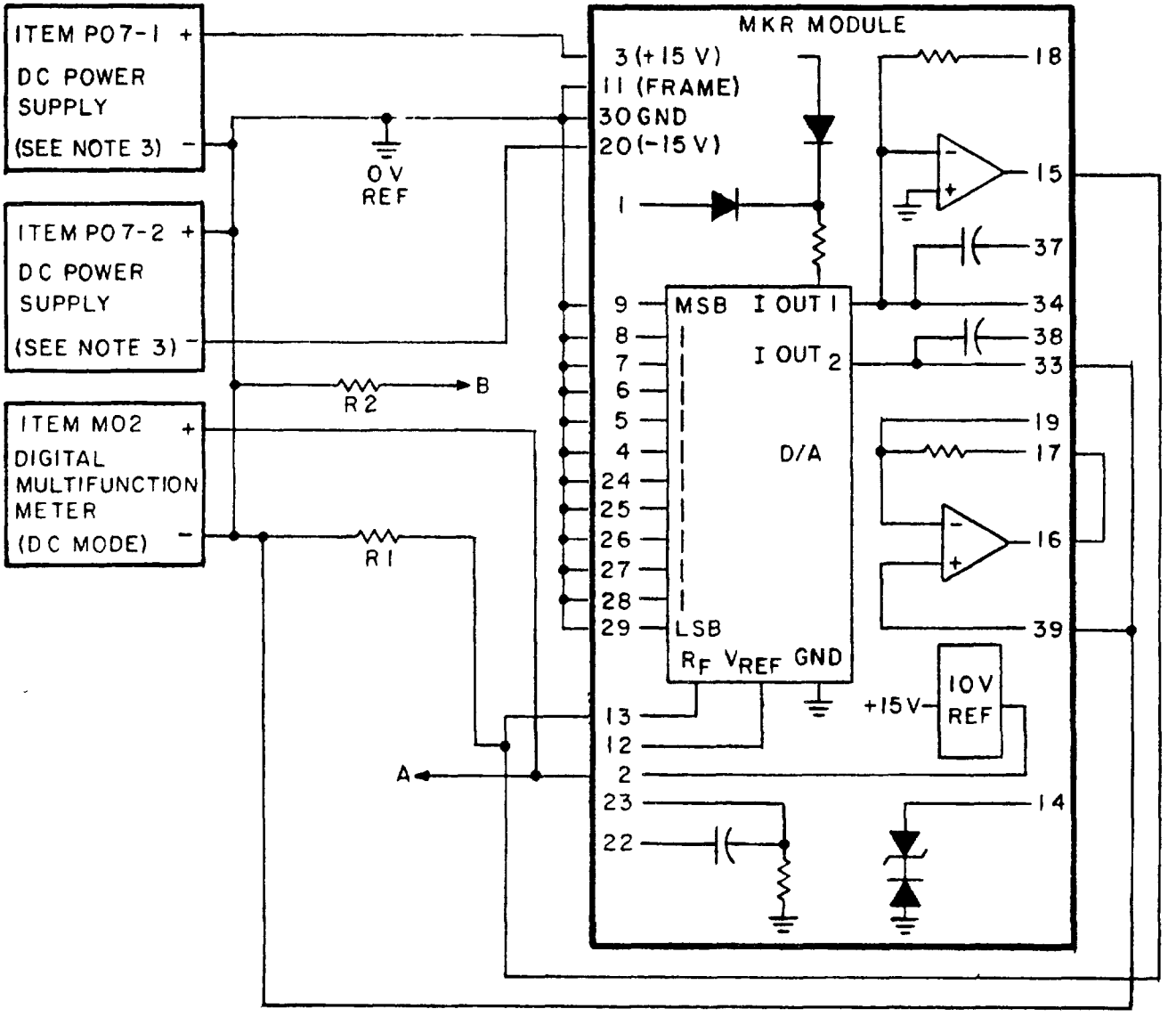
FIGURE 7. Power supply current test circuit for table VI.

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TABLE VII. Reference voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	Reference voltage, voltage regulation	12	9.969	10.031	V	8
D2	Reference voltage, voltage regulation	12	9.969	10.031	V	8
D3	Reference voltage, load regulation	B	9.966	10.031	V	8

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. R2 = 2 kΩ ±1%, 1/8 W.
3. Adjust items P07-1 and P07-2 as follows.
   13.5 V ±1% for test condition D1
   16.5 V ±1% for test condition D2.
   15.0 V ±1% for test condition D3.

FIGURE 8. Reference voltage test circuit for table VII

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TABLE VIII. Offset voltage (op amp) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
E	Offset voltage, op amp	-5	5	mV	9

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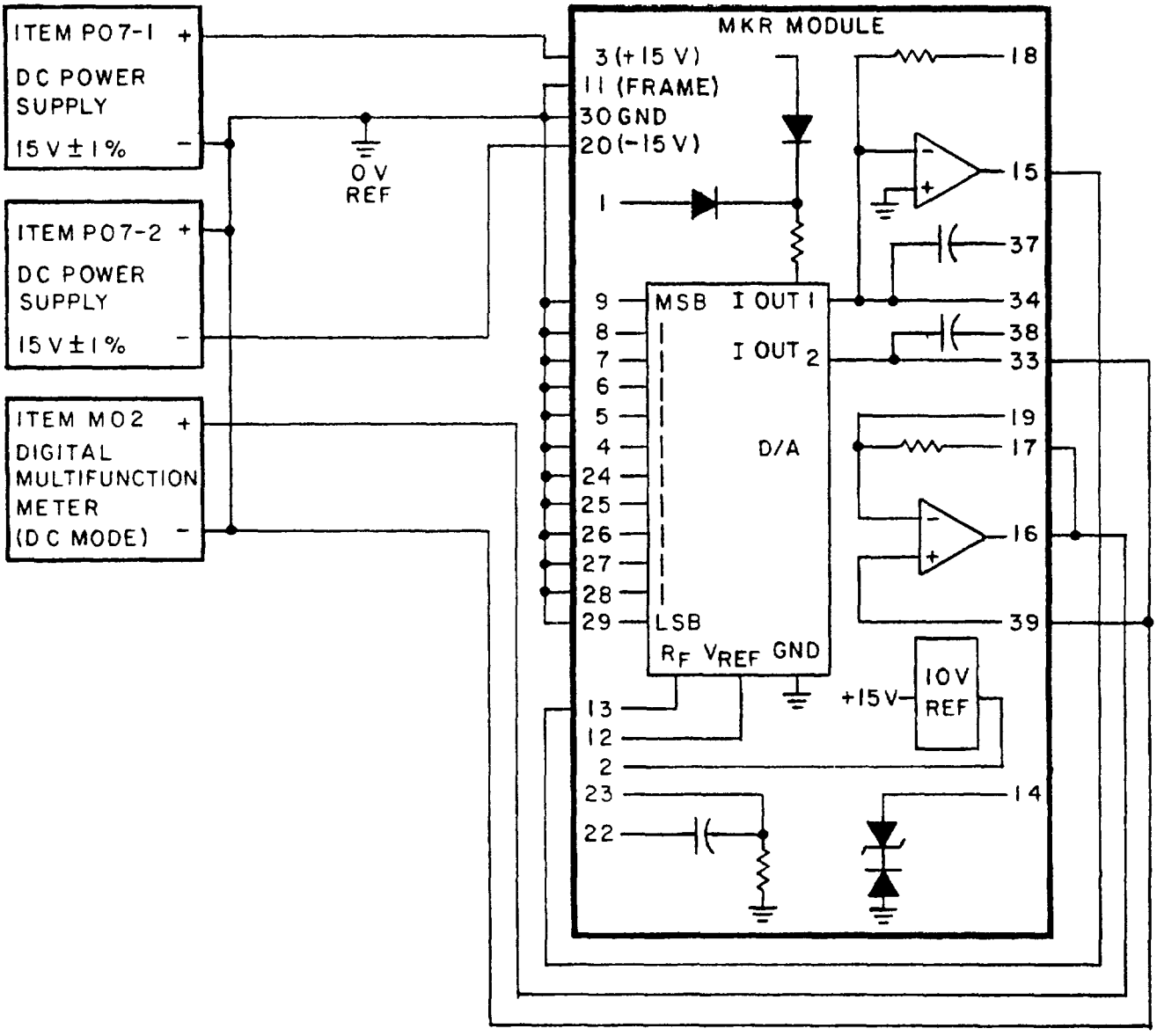


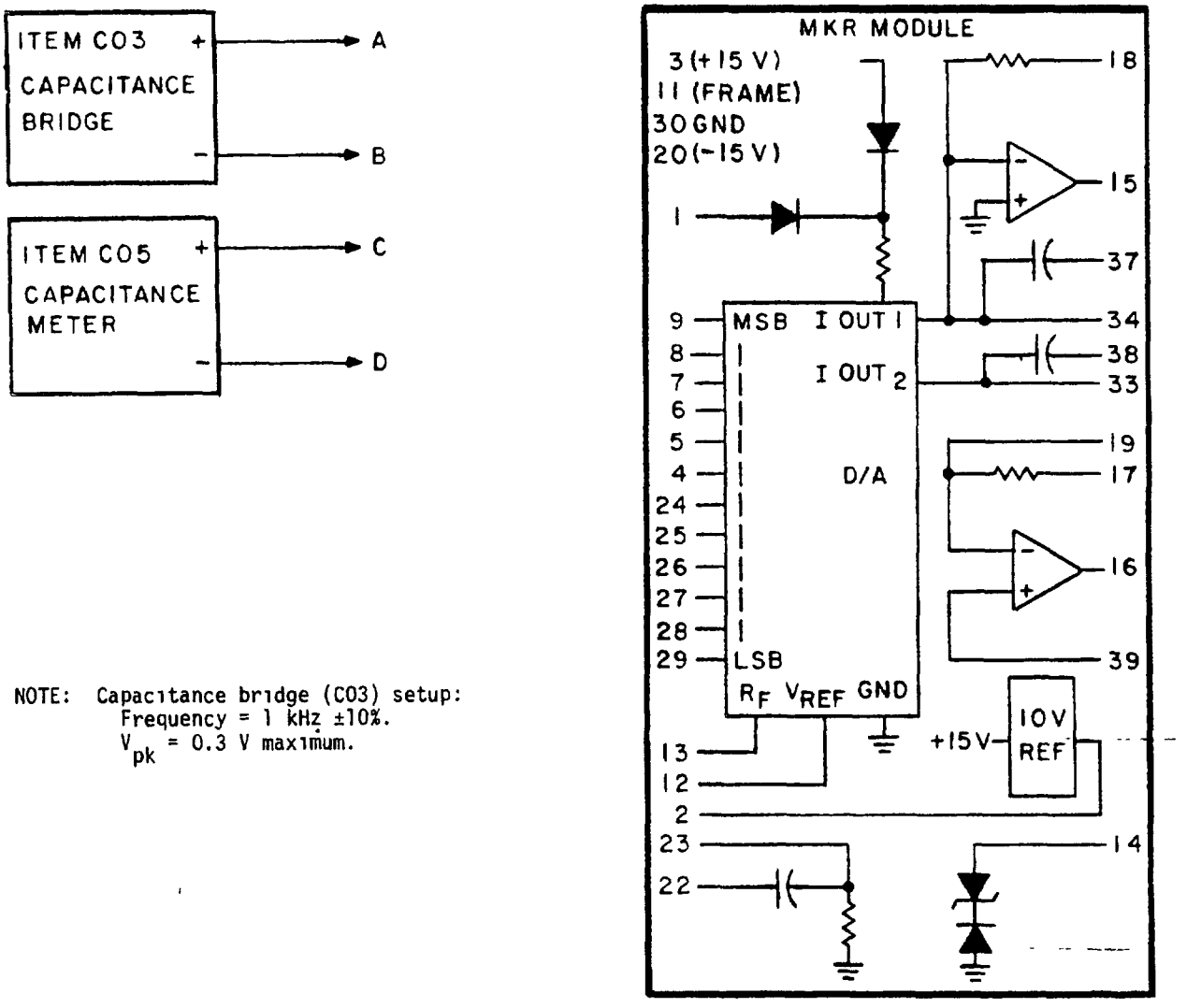
FIGURE 9. Offset voltage (op amp) test circuit for table VIII

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TABLE IX. Capacitance test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	Capacitance, power supply (+15 V)	3	30	Open	Open	0.075	0.125	µF	16
F2	Capacitance, power supply (-15 V)	30	20	Open	Open	0.075	0.125	µF	
F3	Capacitance, C3	22	23	Open	Open	0.089	0.111	µF	
F4	Capacitance, C1	Open	Open	34	37	19	30	pF	
F5	Capacitance, C2	Open	Open	33	38	19	30	pF	

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NOTE: Capacitance bridge (CO3) setup:
Frequency = 1 kHz ±10%.
V_{pk} = 0.3 V maximum.

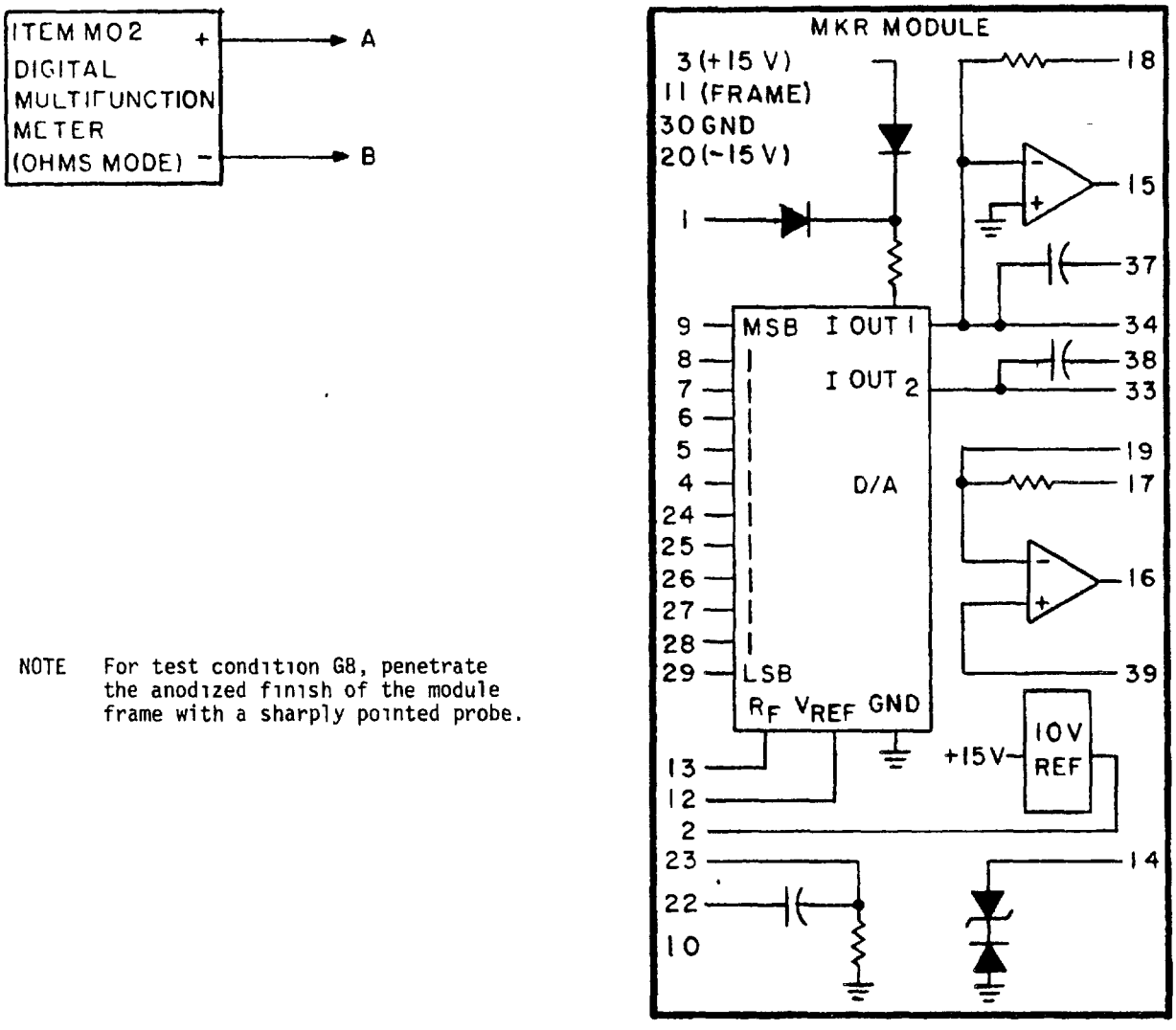
FIGURE 10. Capacitance test circuit for table IX.

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TABLE X. Resistance and continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Resistance, R3	23	30	317	331	kΩ	11
G2	Continuity	21	↓	0	1	Ω	↓
G3	↓	31		↓	↓	↓	
G4		32					
G5		35					
G6		36					
G7		40					
G8		11	Frame				

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NOTE For test condition G8, penetrate the anodized finish of the module frame with a sharply pointed probe.

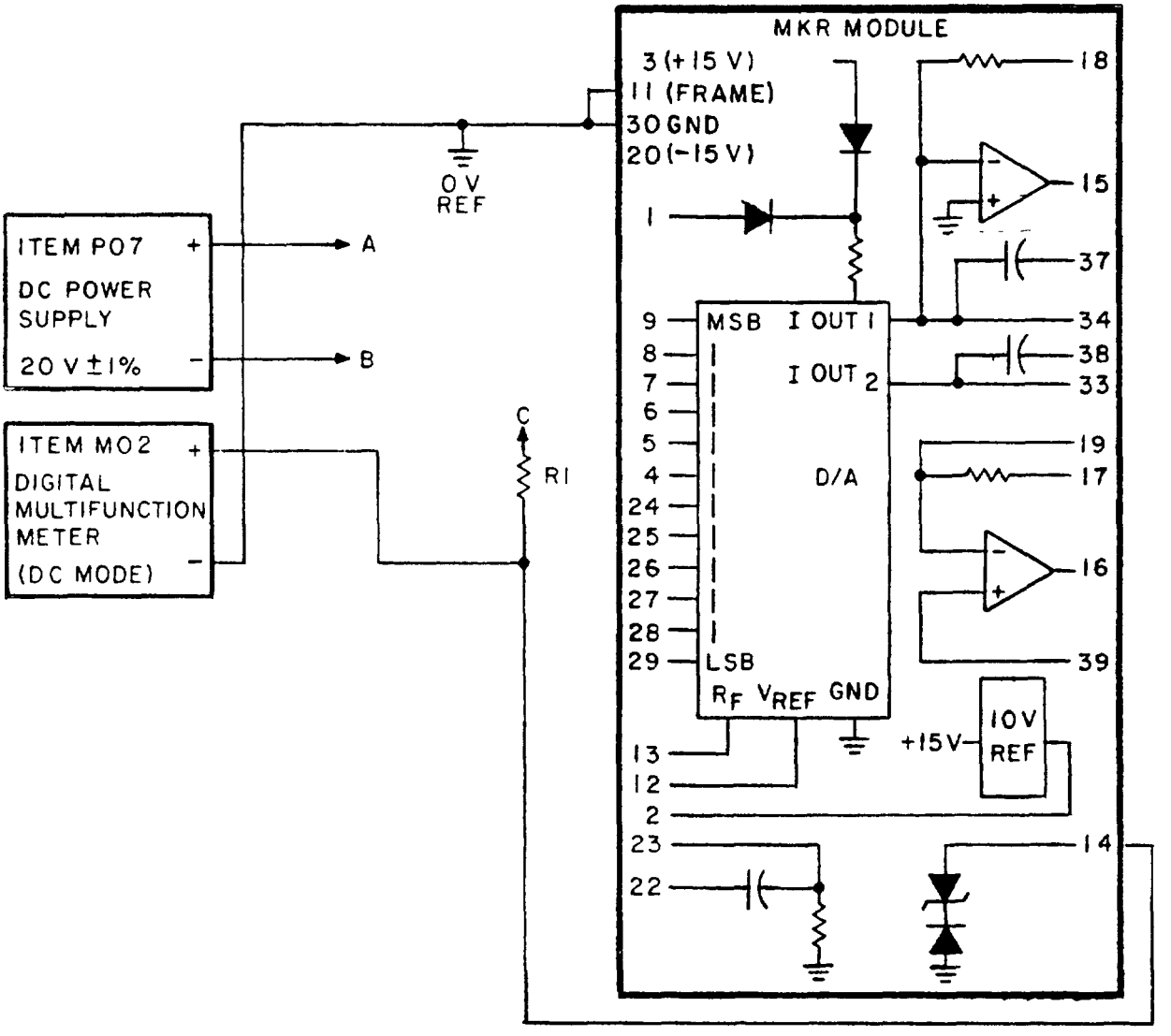
FIGURE 11. Resistance and continuity test circuit for table X.

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TABLE XI. Diode network (zener voltage and reverse bias) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Diode network, zener voltage	30	C	-9.5	-11.3	V	12
H2	Diode network, diode reverse bias	C	30	19.8	20.2	V	12

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NOTE: R1 = 3.01 kΩ ±1%, 1/8 W.

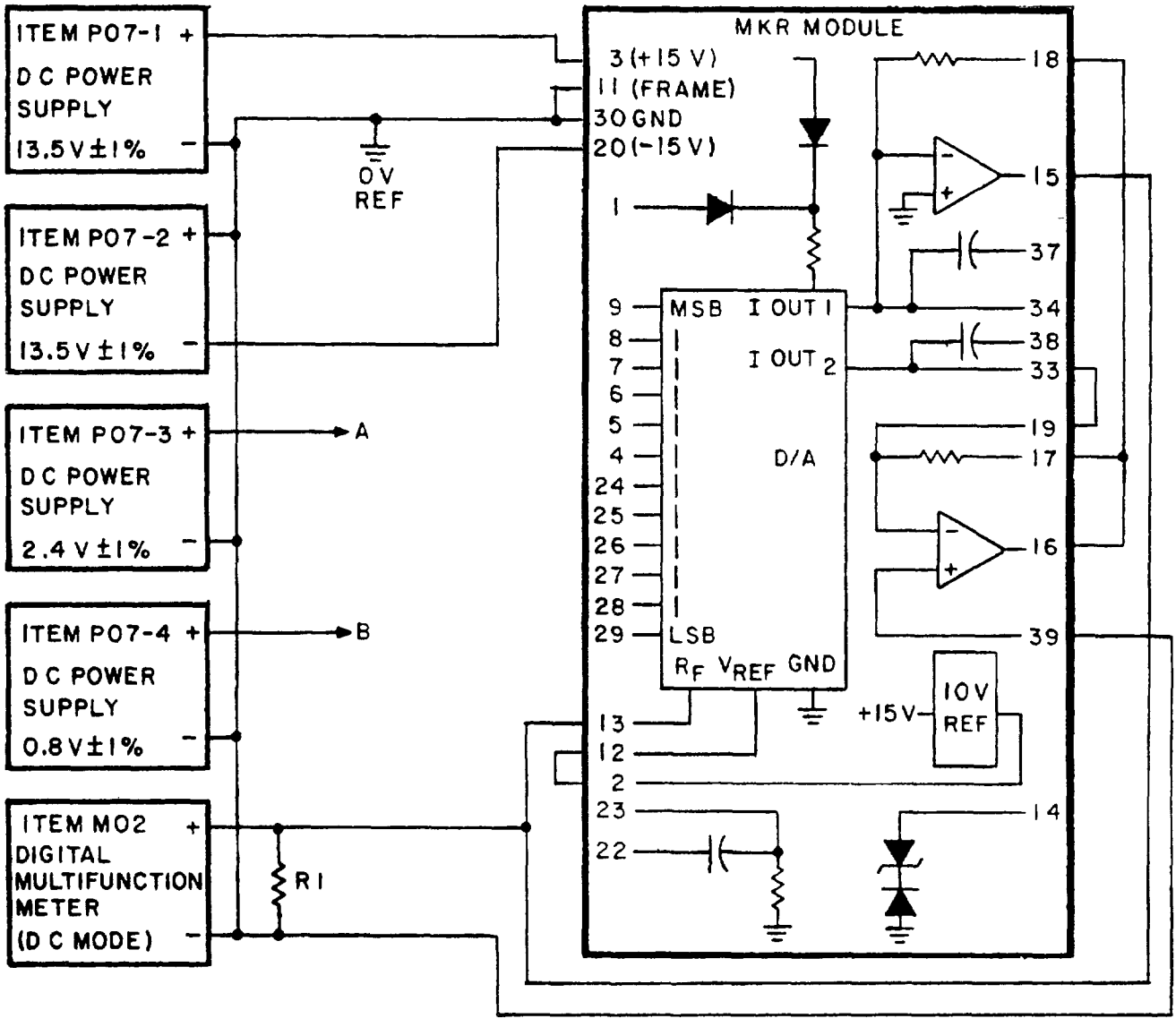
FIGURE 12. Diode network (zener voltage and diode reverse bias) test circuit for table XI.

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TABLE XII. Bipolar accuracy test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Bipolar accuracy, positive full scale	Open	4,5,6,7,8,9,24,25,26,27,28,29	9.72	10.27	V	13
J2	Bipolar accuracy, zero scale	9	4,5,6,7,8,9,24,25,26,27,28,29	-0.14	0.14	V	13
J3	Bipolar accuracy, negative full scale	4,5,6,7,8,9,24,25,26,27,28,29	Open	-9.93	-10.07	V	13

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NOTE: R1 = 10 kΩ ±1%, 1/8 W.

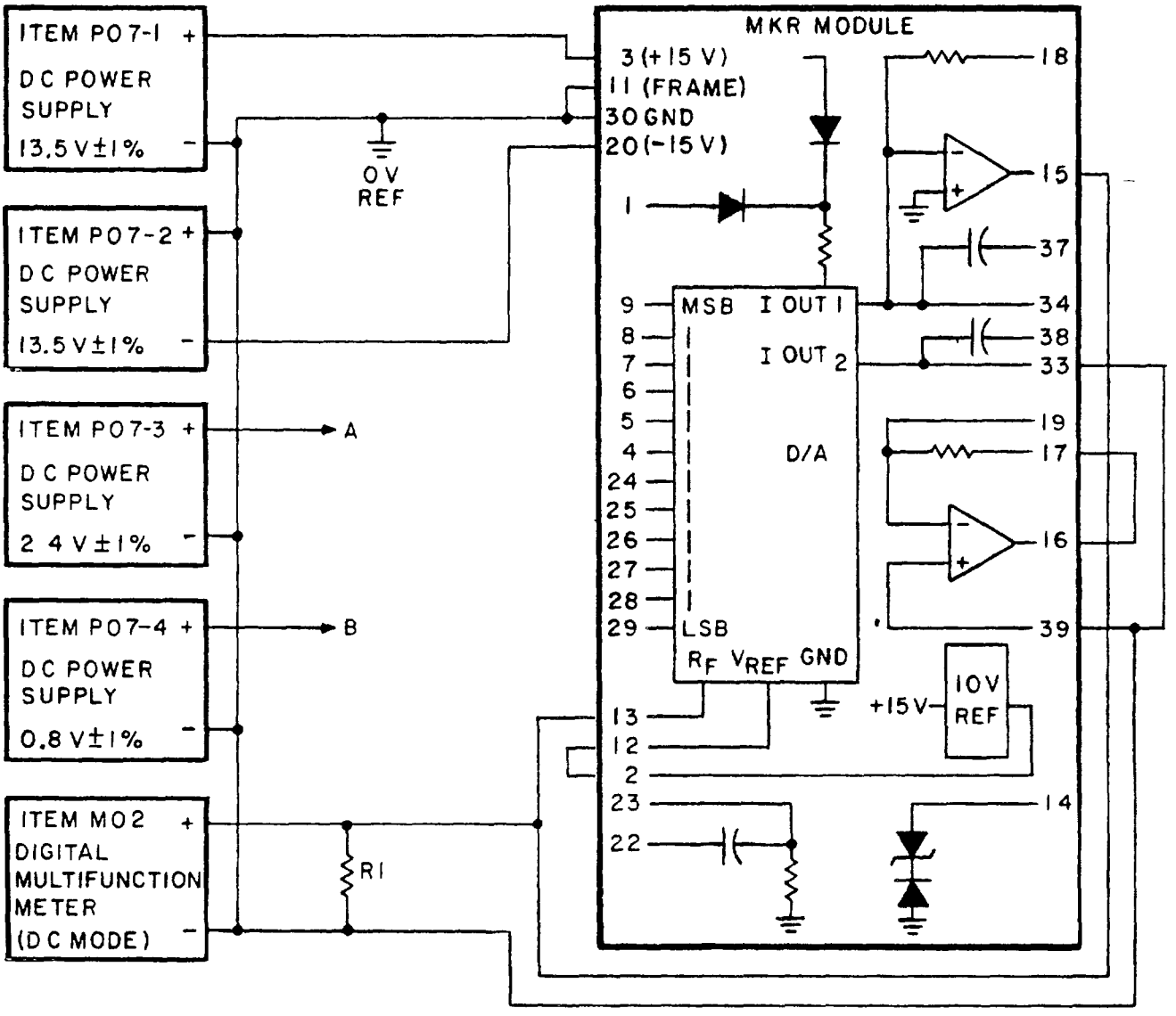
FIGURE 13. Bipolar accuracy test circuit for table XII.

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TABLE XIII. DAC unipolar (offset and full scale error) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	DAC unipolar offset voltage	Open	4,5,6,7,8,9,24,25,26,27,28,29	-5	5	mV	14
K2	DAC unipolar full scale error	4,5,6,7,8,9,24,25,26,27,28,29	Open	-9.93	-10.07	V	14

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NOTE. R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 14. DAC unipolar (offset and full scale error) test circuit for table XIII

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TABLE XIV. DAC linearity check test. 1/ - 3/

Test condition	Parameter	Input code (for reference only)			Connect		Initial limits 25°C±5°C ambient		Unit	Test figure
					A to pin	B to pin	Min	Max		
L1	Bit 12 DAC linearity check	0000	0000	0001	29	4,5,6,7,8,9,24,25, 26,27,28	0.4	4.4	mV	15
L2	Bit 11	0000	0000	0010	28	4,5,6,7,8,9,24,25, 26,27,29	2.9	6.9		
L3	Bit 10	0000	0000	0100	27	4,5,6,7,8,9,24,25, 26,28,29	7.8	11.8		
L4	Bit 9	0000	0000	1000	26	4,5,6,7,8,9,24,25, 27,28,29	17.5	21.5		
L5	Bit 8	0000	0001	0000	25	4,5,6,7,8,9,24,26, 27,28,29	37.1	41.1		
L6	Bit 7	0000	0010	0000	24	4,5,6,7,8,9,25,26, 27,28,29	76.1	80.1		
L7	Bit 6	0000	0100	0000	4	5,6,7,8,9,24,25,26, 27,28,29	154.3	158.3		
L8	Bit 5	0000	1000	0000	5	4,6,7,8,9,24,25,26, 27,28,29	310.5	314.5		
L9	1/16 FSR	0000	1111	1111	4,5,24,25,26, 27,28,29	6,7,8,9	620.6	624.6		
L10	1/16 FSR	0001	0000	0000	6	4,5,7,8,9,24,25,26, 27,28,29	623.0	627.0		
L11	1/8 FSR	0001	1111	1111	4,5,6,24,25, 26,27,28,29	7,8,9	1.2456	1.2496	V	
L12	1/8 FSR	0010	0000	0000	7	4,5,6,8,9,24,25,26, 27,28,29	1.2480	1.2520		
L13	3/16 FSR	0010	1111	1111	4,5,7,24,25, 26,27,28,29	6,8,9	1.8706	1.8746		
L14	3/16 FSR	0011	0000	0000	6,7	4,5,8,9,24,25,26, 27,28,29	1.8730	1.8770		
L15	1/4 FSR	0011	1111	1111	4,5,6,7,24,25, 26,27,28,29	8,9	2.4976	2.4996		

See footnotes at end of table.

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TABLE XIV. DAC linearity check test - Continued. 1/ - 3/

Test condition	Parameter	Input code (for reference only)			Connect		Initial limits 25°C±5°C ambient		Unit	Test figure
					A to pin	B to pin	Min	Max		
L16	1/4 FSR DAC linearity check	0100	0000	0000	8	4,5,6,7,9,24,25,26, 27,28,29	2.4980	2.5020	V	15
L17	5/16 FSR	0100	1111	1111	4,5,8,24,25, 26,27,28,29	6,7,9	3.1206	3.1246		
L18	5/16 FSR	0101	0000	0000	6,8	4,5,7,9,24,25,26,27, 28,29	3.1230	3.1270		
L19	3/8 FSR	0101	1111	1111	4,5,6,8,24,25, 26,27,28,29	7,9	3.7456	3.7496		
L20	3/8 FSR	0110	0000	0000	7,8	4,5,6,9,24,25,26, 27,28,29	3.7480	3.7520		
L21	7/16 FSR	0110	1111	1111	4,5,7,8,24,25, 26,27,28,29	6,9	4.3706	4.3746		
L22	7/16 FSR	0111	0000	0000	6,7,8	4,5,9,24,25,26,27, 28,29	4.3730	4.3770		
L23	1/2 FSR	0111	1111	1111	4,5,6,7,8,24, 25,26,27,28,29	9	4.9956	4.9996		
L24	1/2 FSR	1000	0000	0000	9	4,5,6,7,8,24,25,26, 27,28,29	4.9980	5.0020		
L25	9/16 FSR	1000	1111	1111	4,5,9,24,25, 26,27,28,29	6,7,8	5.6206	5.6246		
L26	9/16 FSR	1001	0000	0000	6,9	4,5,7,8,24,25,26,27, 28,29	5.6230	5.6270		
L27	5/8 FSR	1001	1111	1111	4,5,6,9,24,25, 26,27,28,29	7,8	6.2456	6.2496		
L28	5/8 FSR	1010	0000	0000	7,9	4,5,6,8,24,25,26,27, 28,29	6.2480	6.2520		
L29	11/16 FSR	1010	1111	1111	4,5,7,9,24,25, 26,27,28,29	6,8	6.8706	6.8746		

See footnotes at end of the table.

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TABLE XIV. DAC linearity check test - Continued. 1/ - 3/

Test condition	Parameter	Input code (for reference only)			Connect		Initial limits 25°C±5°C ambient		Unit	Test figure
					A to pin	B to pin	Min	Max		
L30	11/16 FSR DAC linearity check	1011	0000	0000	6,7,9	4,5,8,24,25,26,27,28,29	6.8730	6.8770	V	15
L31	3/4 FSR	1011	1111	1111	4,5,6,7,9,24,25,26,27,28,29	8	7.4956	7.4996		
L32	3/4 FSR	1100	0000	0000	8,9	4,5,6,7,24,25,26,27,28,29	7.4980	7.5020		
L33	13/16 FSR	1100	1111	1111	4,5,8,9,24,25,26,27,28,29	6,7	8.1206	8.1246		
L34	13/16 FSR	1101	0000	0000	6,8,9	4,5,7,24,25,26,27,28,29	8.1230	8.1270		
L35	7/8 FSR	1101	1111	1111	4,5,6,8,9,24,25,26,27,28,29	7	8.7456	8.7496		
L36	7/8 FSR	1110	0000	0000	7,8,9	4,5,6,24,25,26,27,28,29	8.7480	8.7520		
L37	15/16 FSR	1110	1111	1111	4,5,7,8,9,24,25,26,27,28,29	6	9.3706	9.3746		
L38	15/16 FSR	1111	0000	0000	6,7,8,9	4,5,24,25,26,27,28,29	9.3730	9.3770		

1/ FSR - full scale range

2/ The output voltage as measured by item M02 is actually a negative voltage, however, the item M02 leads are reversed so as to display a positive voltage.

3/ This test measures DAC linearity where ±2.4 mV limit = ±1 LSB.

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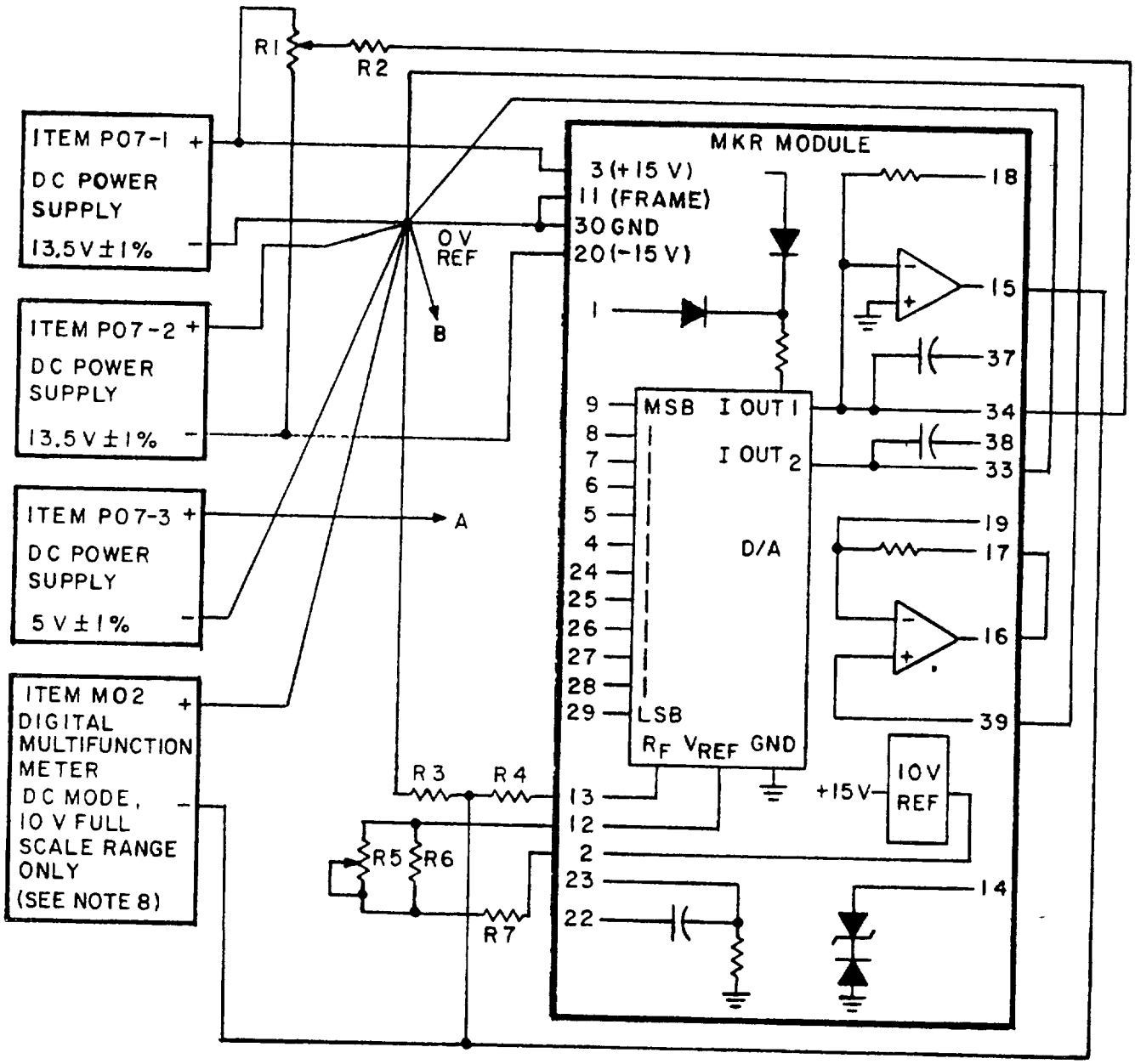


FIGURE 15. DAC linearity check test circuit for table XIV.

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NOTES:
1. R1 = 50 kΩ ±10%, 1/4 W multi-turn potentiometer.
2. R2 = 1 MΩ ±10%, 1/8 W.
3. R3 = 10 kΩ ±1%, 1/8 W.
4. R4 = 1 kΩ ±1%, 1/8 W.
5. R5 = 500Ω ±10%, 1/4 W, multi-turn potentiometer.
6. R6 = 499Ω ±1%, 1/8 W.
7. R7 = 866Ω ±1%, 1/8 W.
8. The item M02 used must have an accuracy of ±0.005% of reading ±0.001% of range or better on the 10 V full scale range. Not all item M02 listed in MIL-STD-1665 will meet this requirement.
9. Prior to making measurements, adjust out the offset and full scale errors as follows (keep item M02 set to the 10 V full scale range):

   a. Connect point B to pins 4-9 and 24-29.
   b. Allow the output time to stabilize and adjust R1 until item M02 reads 0 ±0.1 mV.
   c. Disconnect point B from the indicated pins and connect point A to pins 4-9 and 24-29.
   d. Adjust R5 until item M02 reads 9.9976 ±0.0002.
   e. Disconnect point A.

10. All measurements for this test condition shall be made without interruption to eliminate errors due to test equipment and module drift.
11. All 0 V reference connections shall be made at one point close to the module pin 30 as shown.

FIGURE 15. DAC linearity check test circuit for table XIV - Continued.

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TABLE XV. DAC differential linearity error check test. 1/ - 4/

Test condition	Parameter	Calculation	Initial limits 25°±5°C ambient		Unit
			Min	Max	
M1	1/16 FSR DAC differential linearity error check	L10-L9	0.0	4.9	mV
M2	1/8 FSR	L12-L11			
M3	3/16 FSR	L14-L13			
M4	1/4 FSR	L16-L15			
M5	5/16 FSR	L18-L17			
M6	3/8 FSR	L20-L19			
M7	7/16 FSR	L22-L21			
M8	1/2 FSR	L24-L23			
M9	9/16 FSR	L26-L25			
M10	5/8 FSR	L28-L27			
M11	11/16 FSR	L30-L29			
M12	3/4 FSR	L32-L31			
M13	13/16 FSR	L34-L33			
M14	7/8 FSR	L36-L35			
M15	15/16 FSR	L38-L37			

1/ FSR - full scale range.

2/ Test conditions M1 through M15 are calculations. The result of subtracting the indicated test condition readings shall meet the limits shown.

3/ This test condition verifies DAC monotonicity (differential linearity error ≤ ±1 LSB) only for the tested codes.

4/ Differential linearity error (table I) = $\frac{\text{Test condition M} - 2.45 \text{ mV}}{2.45 \text{ mV}}$.

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TABLE XVI. Gain frequency response check test. 1/

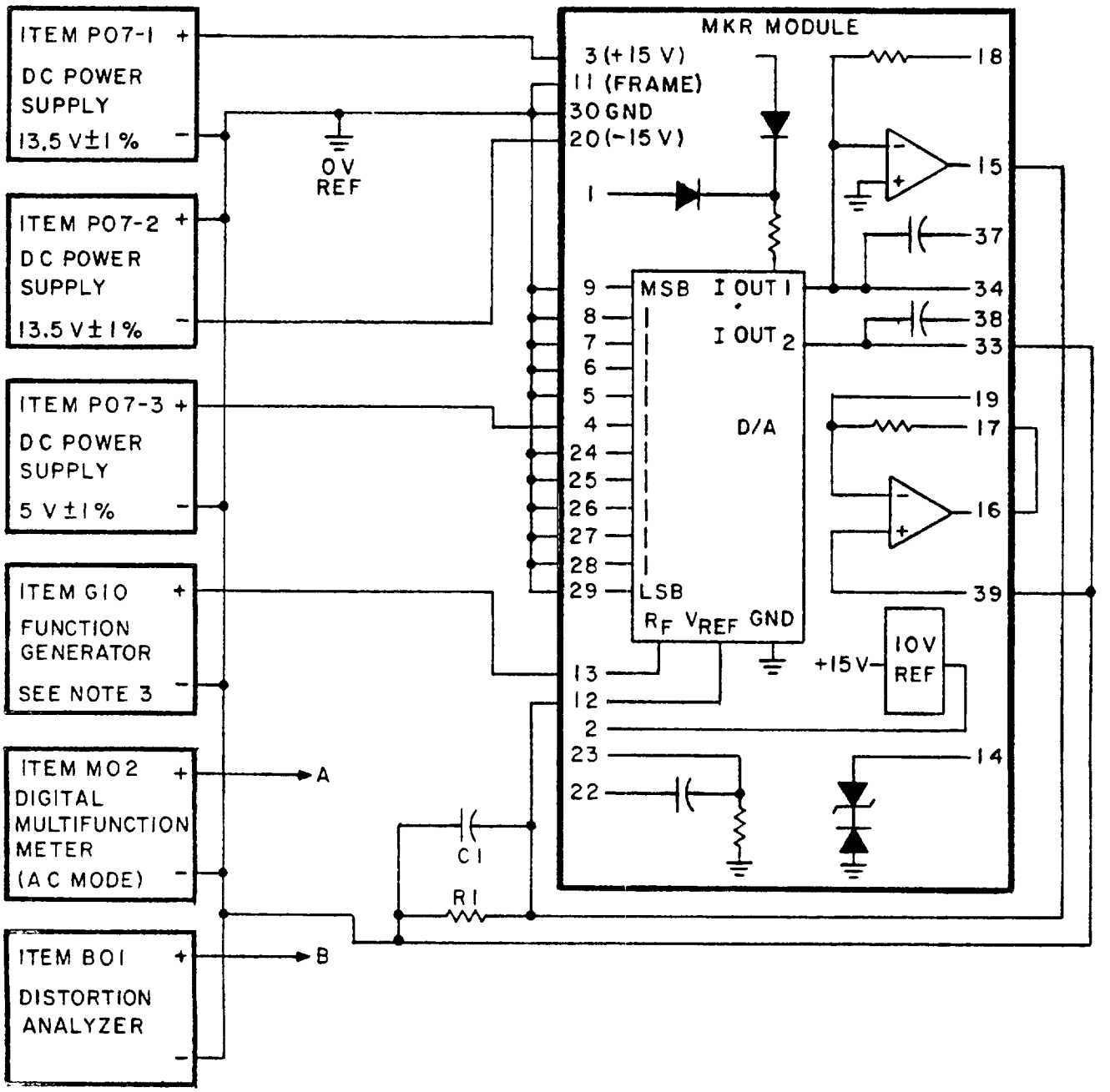
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Gain frequency response check (10 kHz)	15	Open	1.808	1.982	Vrms	16
N2	Gain frequency response check (30 kHz)	15	Open	0.865	1.982	Vrms	16
N3	Gain frequency response check (100 kHz)	15	Open	0.498	1.982	Vrms	16

1/ Gain (dB) = 20 log $\frac{\text{Item M02 reading}}{0.030\ \text{V}}$

TABLE XVII. Distortion test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P	Distortion	Open	15	0.0	0.5	%	16

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. C1 = 100 pF ±10%.
3. Item G10 setup:
   Sine wave amplitude = 0.30 Vrms ±1% (recheck after frequency change).
   Frequency = 10 kHz ±1% for test condition N1.
   = 30 kHz ±1% for test condition N2.
   = 100 kHz ±1% for test condition N3 and P.

FIGURE 16. Gain-frequency response check and distortion test circuit for tables XVI and XVII.

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TABLE XVIII. V_{REF} input impedance (low-level and high-level) check test. 1/

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	V_{REF} input impedance check, low level	Open	4,5,6,7,8,9,24, 25,26,27,28,29	0.5	2.0	mA	17
Q2	V_{REF} input impedance check, high level	4,5,6,7,8,9,24,25, 26,27,28,29	Open	0.5	2.0	mA	17

1/ Input impedance = $\frac{10\ V}{\text{Item X05 reading}}$

TABLE XIX. Settling time (t_{SHL} and t_{SLH}) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
R1	0 to -10 V Settling time (t_{SHL})	0.0	5.5	µs	18,19
R2	-10 to 0 V Settling time(t_{SLH})	0.0	5.5	µs	18,19

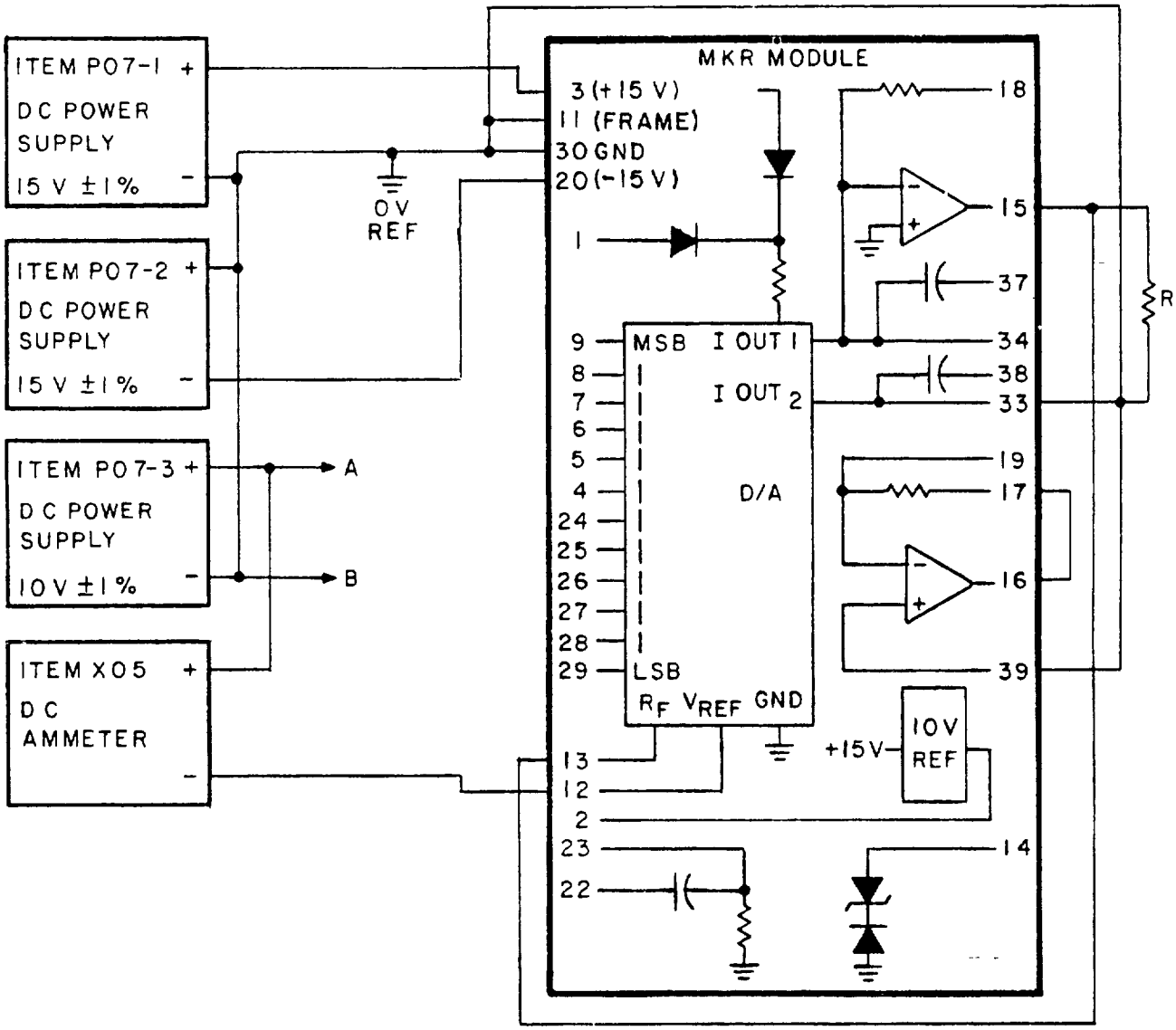
TABLE XX. Power supply rejection ratio check test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply rejection ratio check 1/	0.0	8.9	mVrms	20

1/ This test verifies a power supply rejection ratio of 40 dB.

$$PSRR\ (dB) = 20 \log \frac{\text{Item W04 reading.}}{0.71\ Vrms}$$

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W
2. This test verifies a V_{REF} input impedance of 5 to 20 kΩ.

FIGURE 17. V_{REF} input impedance (low-level and high-level) check test circuit for table XVIII

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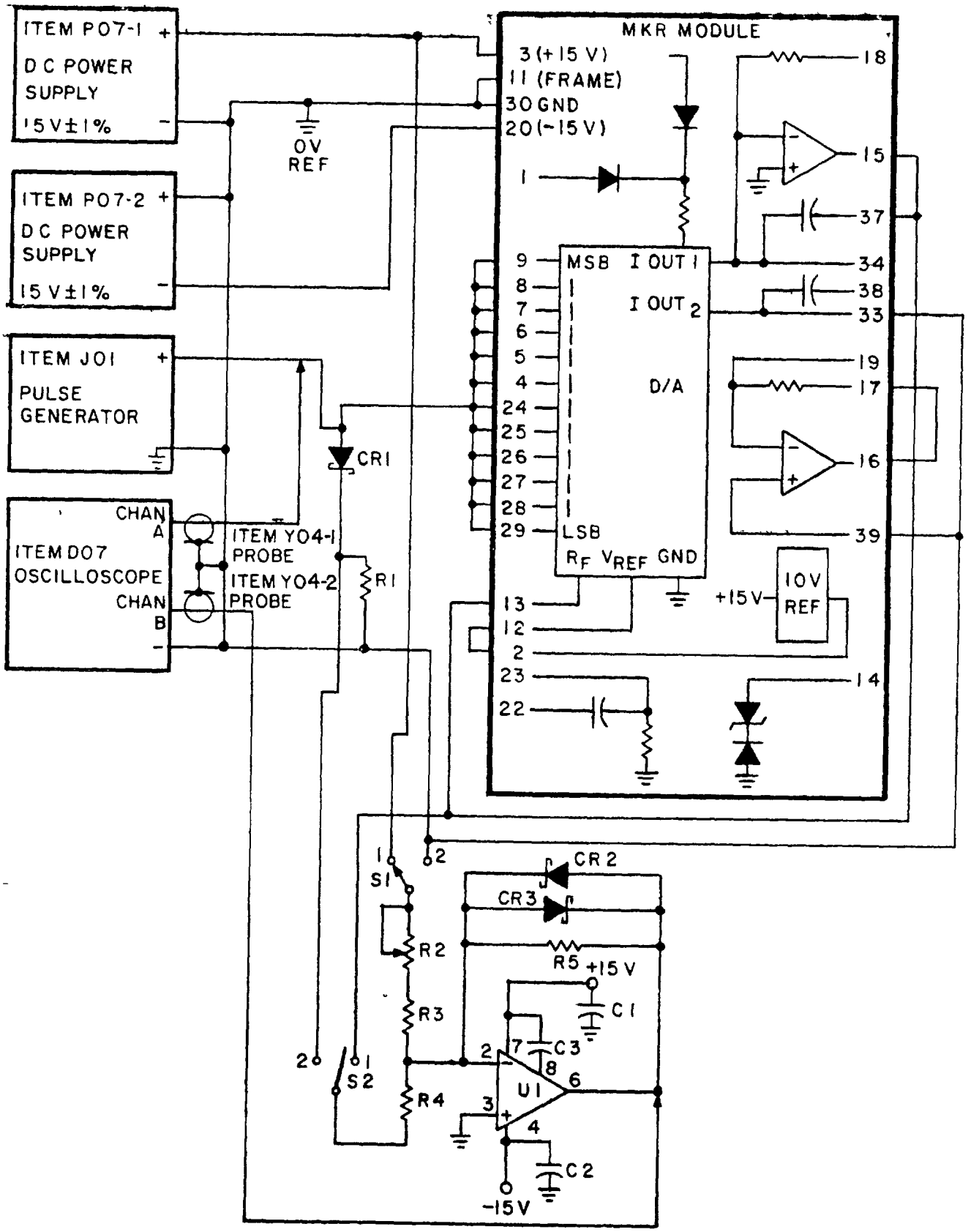


FIGURE 18. Settling time (t_{SHL} and t_{SLH}) test circuit for table XIX.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 500Ω ±5%, 1/4 W multi-turn potentiometer.
3. R3 = 2.74 kΩ ±1%, 1/8 W.
4. R4 = 2 kΩ ±1%, 1/8 W.
5. R5 = 20 kΩ ±1%, 1/8 W.
6. CR1-CR3 = Schottky diodes 1N5165, or equivalent.
7. C1 = C2 = 0.1 µF ±20%, 50 V.
8. C3 = 15 pF ±20%.
9. U1 = Harris HA2515 or equivalent.
10. S1, S2 = SPDT switch.
11. When performing test condition R on a new setup, check the measuring circuit settling time as follows:

    a. Set switches S1 and S2 to positive 2 (disregard potentiometer R2 adjustment for this check).
    b. Measure the time designated by t_{SLH} on figure 19. This should be less than 0.5 µs. If the measurement is greater than this, correct the measuring setup before proceeding. Some possible sources of excess time are U1 too slow, excess lead length or excess stray capacitance.

12. Test condition R1 shall be performed as follows:

    a. Set S1 and S2 to position 1.
    b. Make specified connections and energize the module.
    c. Adjust potentiometer R2 until the U1 output pulse, high level, is 0 ±2 mV.
    d. Measure t_{SHL}.

13. Test condition R2 shall be performed by making the specified connections and measuring t_{SLH} (disregard potentiometer R2 adjustment). Set S1 to position 2 and S2 to position 1.

14. Settling time (t_s) = $\sqrt{(t_s \text{ measured})^2 - (t_s \text{ error})^2}$
    where t_s measured = oscilloscope measurement for test condition R1 and R2
    and t_s error = t_s measured per note 11.

FIGURE 18. Settling time (t_{SHL} and t_{SLH}) test circuit for table XIX - Continued.

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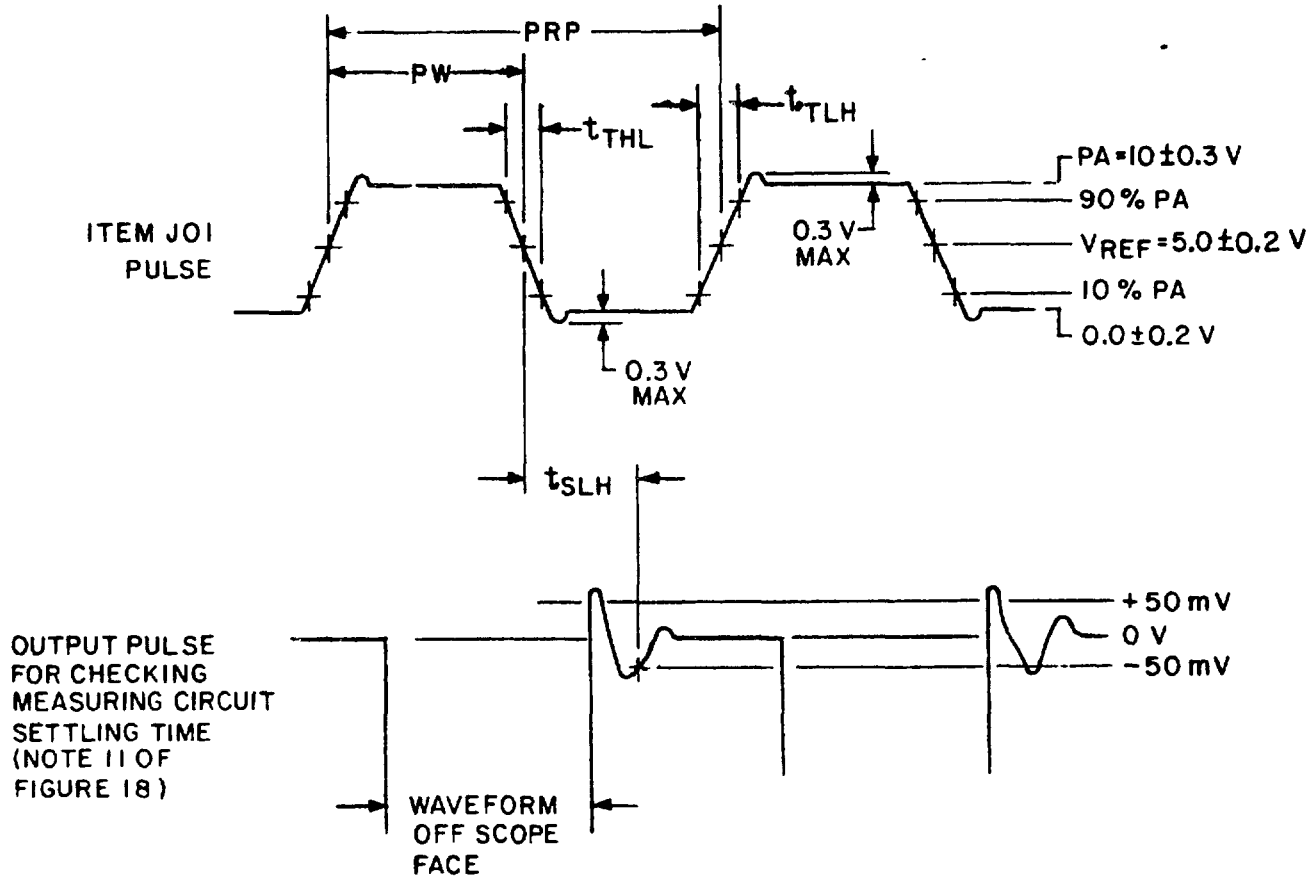
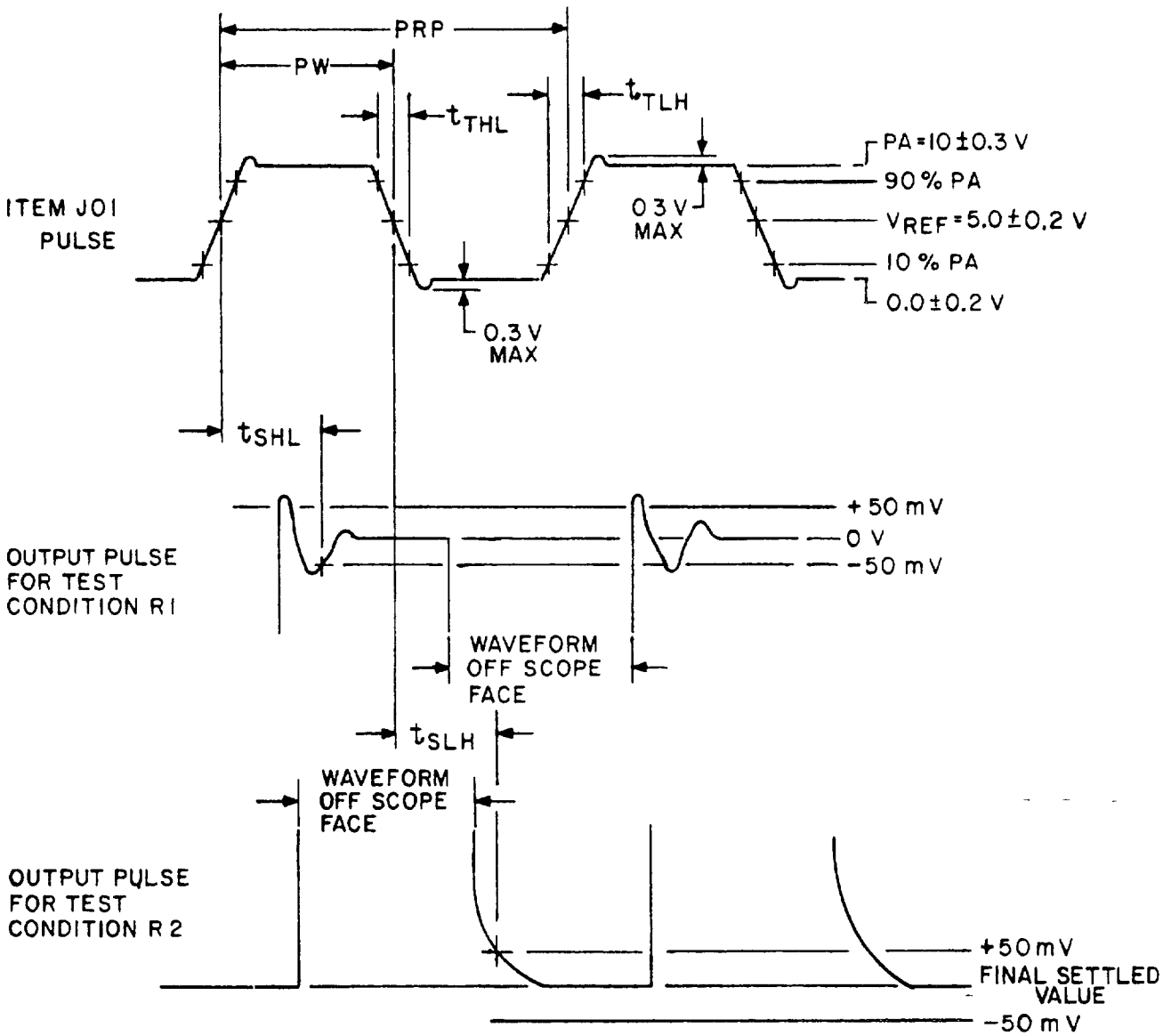


FIGURE 19. Settling time (t_{SHL} and t_{SLH}) test circuit waveforms for table XIX.

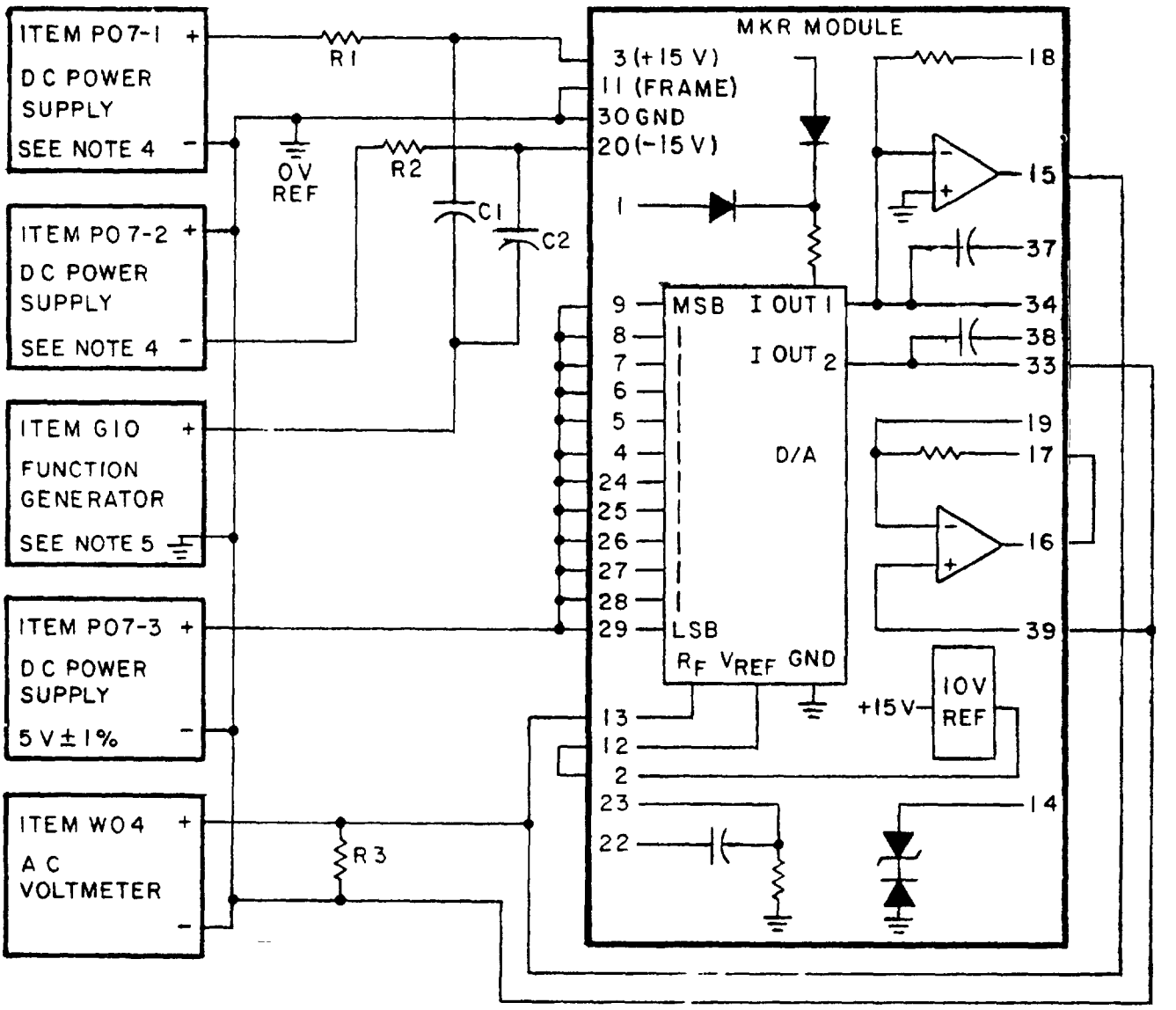
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NOTES:
1. Item J01 pulse parameters:
   PRP = 40 ±1 µs.
   PW = 20 ±0.5 µs.
   $t_{THL} = t_{TLH}$ = 20 ±10 ns.
2. The settling time symbols t_{SHL} and t_{SLH} refer to the signal on pin 15 of the MKR module. The amplifier/clipper U1 inverts this signal before it is displayed by item D07.
3. All voltages are references to 0 V_{REF}.

FIGURE 19. Settling time (t_{SHL} and t_{SLH}) test circuit waveforms for table XIX - Continued.

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NOTES:
1. R1 = R2 = 100Ω ±1%, 1/8 W.
2. R3 = 10 kΩ ±1%, 1/8 W.
3. C1 = C2 = 20 µF ±20%, 20 Vdc electrolytic.
4. Adjust item PO7-1 to obtain +15 V ±1% at pin 3.
   Adjust item PO7-2 to obtain -15 V ±1% at pin 20.
5. Adjust item G10 for 0.71 Vrms ±5%, 1 kHz ±1% sine wave at pins 3 and 20 (use item W04 to make this measurement).

FIGURE 20. Power supply rejection ratio check test circuit for table XX.

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TABLE XXI. AC feedthrough check test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
T	AC feedthrough check 1/	0	10	mVrms	21

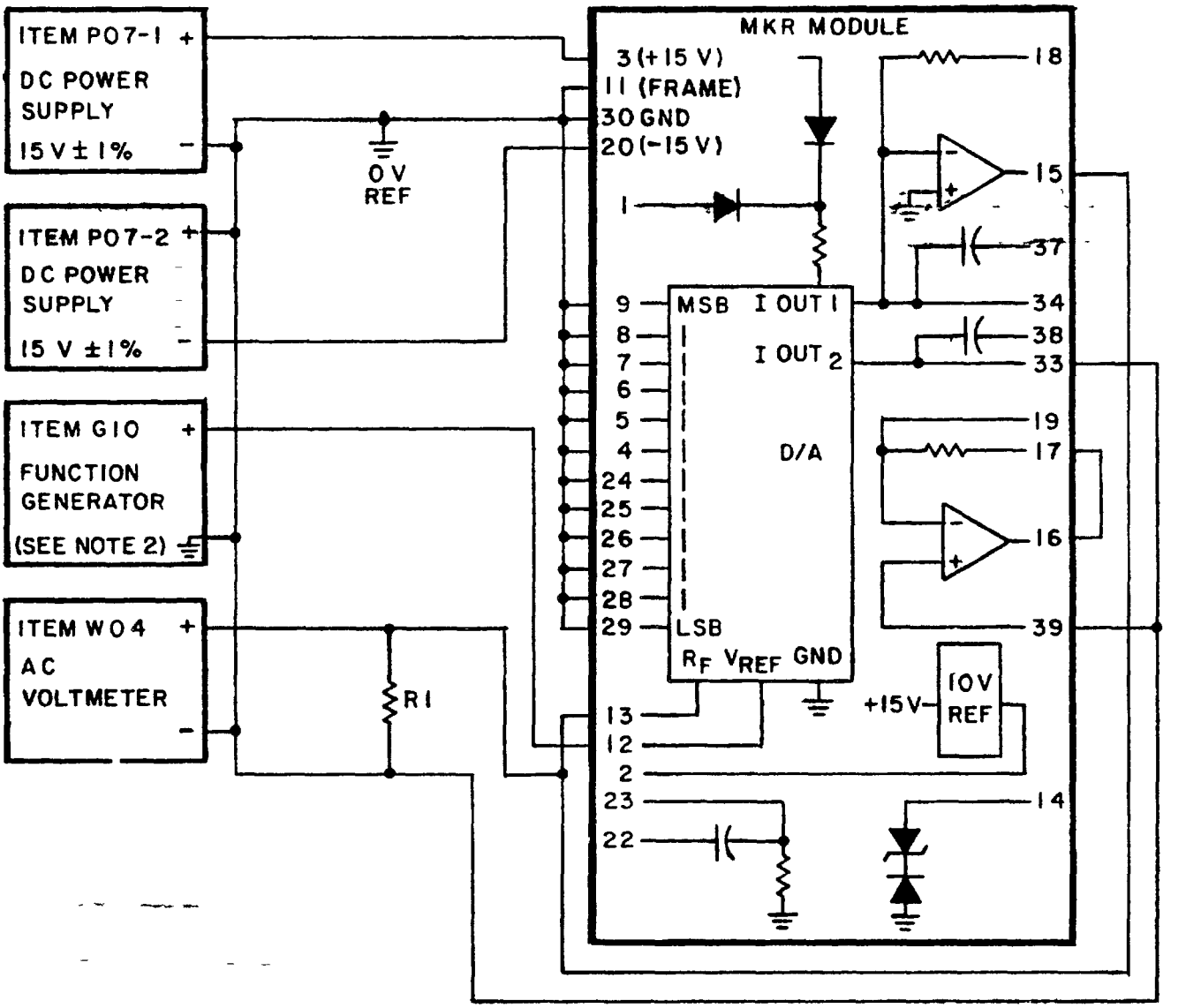
1/ Feedthrough (dB) = 20 log $\frac{\text{Item W04 reading}}{\text{10 Vrms}}$.

TABLE XXII. Noise test. 1/

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
U1	Noise, op amp (20 Hz)	16	-70	-∞	dBV	22
U2	Noise, D/A converter (20 Hz)	23	-50			
U3	Noise, op amp (100 Hz)	16	-80			
U4	Noise, D/A converter (100 Hz)	23	-60			
U5	Noise, op amp (1 kHz)	16	-90			
U6	Noise, D/A converter (1 kHz)	23	-70			
U7	Noise, op amp (10 kHz)	16	-90			
U8	Noise, D/A converter (10 kHz)	23	-70			
U9	Noise, op amp (40 kHz)	16	-90			
U10	Noise, D/A converter (40 kHz)	23	-70			

1/ The circuit under test noise is 40 dB less than recorded on item B03 due to the amplification of item A03. The above limits refer to the average noise recorded by item B03.

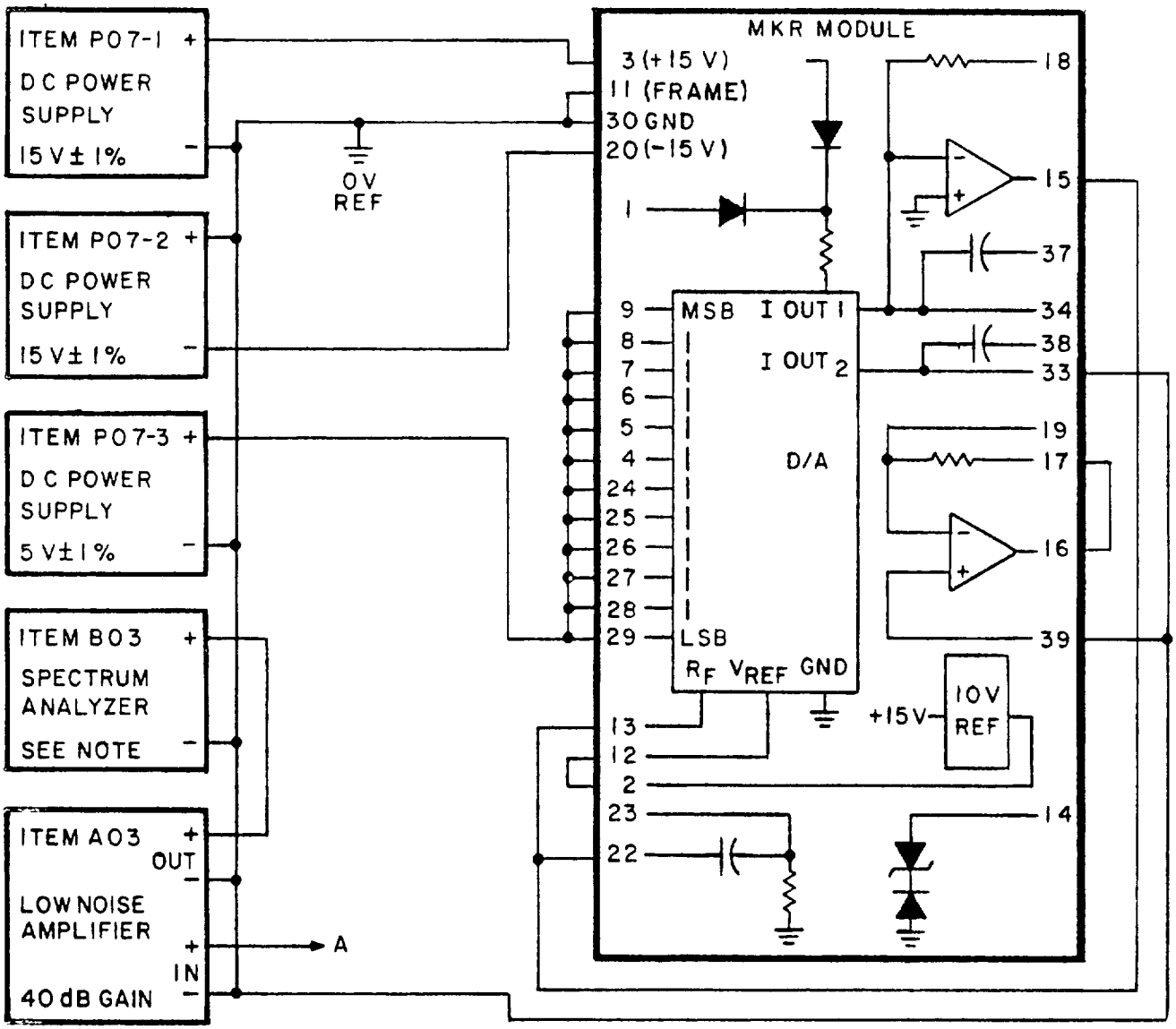
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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. Item G10 setup;
   Sine wave amplitude = 10 Vrms ±1%.
   Frequency = 10 kHz ±1%.

FIGURE 21. AC feedthrough check test circuit for table XXI.

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NOTE: Item B03 setup:
Bandwidth = 10 Hz.
Amplitude mode = log, 10 dB/div.
Frequency = 20 Hz ±1% for test conditions U1, U2.
= 100 Hz ±1% for test conditions U3, U4.
= 1 kHz ±1% for test conditions U5, U6.
= 10 kHz ±1% for test conditions U7, U8.
= 40 kHz ±1% for test conditions U9, U10.
Frequency span/div = 0.
Observe average noise reading and compare to table XXII limits

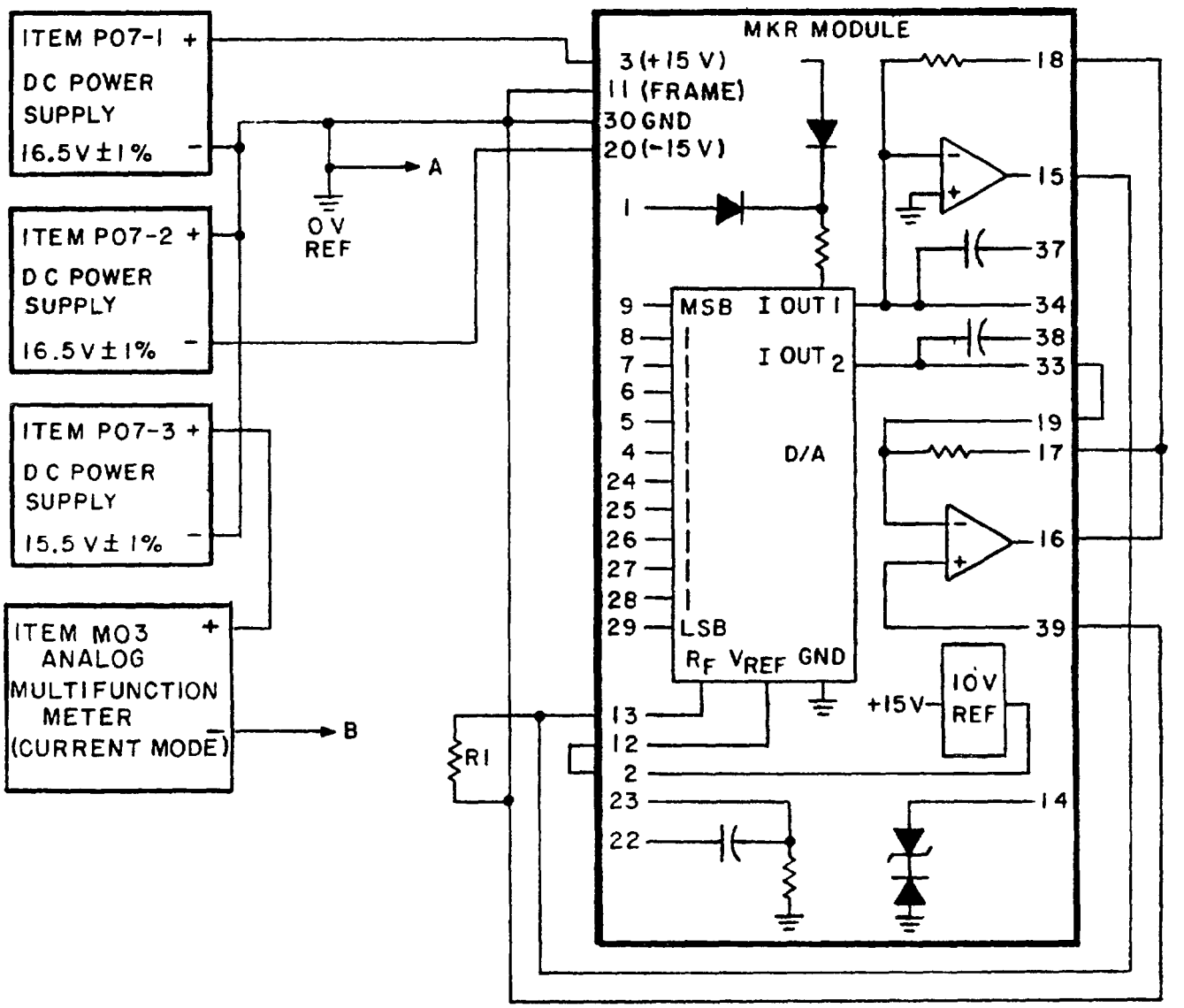
FIGURE 22. Noise test circuit for table XXII.

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TABLE XXIII. Digital input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	Digital input current, high level	5,6,7,8,9,24,25,26,27, 28,29	4	-1	1	µA	23
V2		4,6,7,8,9,24,25,26,27, 28,29	5				
V3		4,5,7,8,9,24,25,26,27, 28,29	6				
V4		4,5,6,8,9,24,25,26,27, 28,29	7				
V5		4,5,6,7,9,24,25,26,27, 28,29	8				
V6		4,5,6,7,8,24,25,26,27, 28,29	9				
V7		4,5,6,7,8,9,25,26,27, 28,29	24				
V8		4,5,6,7,8,9,24,26,27, 28,29	25				
V9		4,5,6,7,8,9,24,25,27, 28,29	26				
V10		4,5,6,7,8,9,24,25,26, 28,29	27				
V11		4,5,6,7,8,9,24,25,26, 27,29	28				
V12		4,5,6,7,8,9,24,25,26, 27,28	29				

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. The digital input voltage must always be at least 1.0 V below the pin 3 voltage to prevent damage to the module.

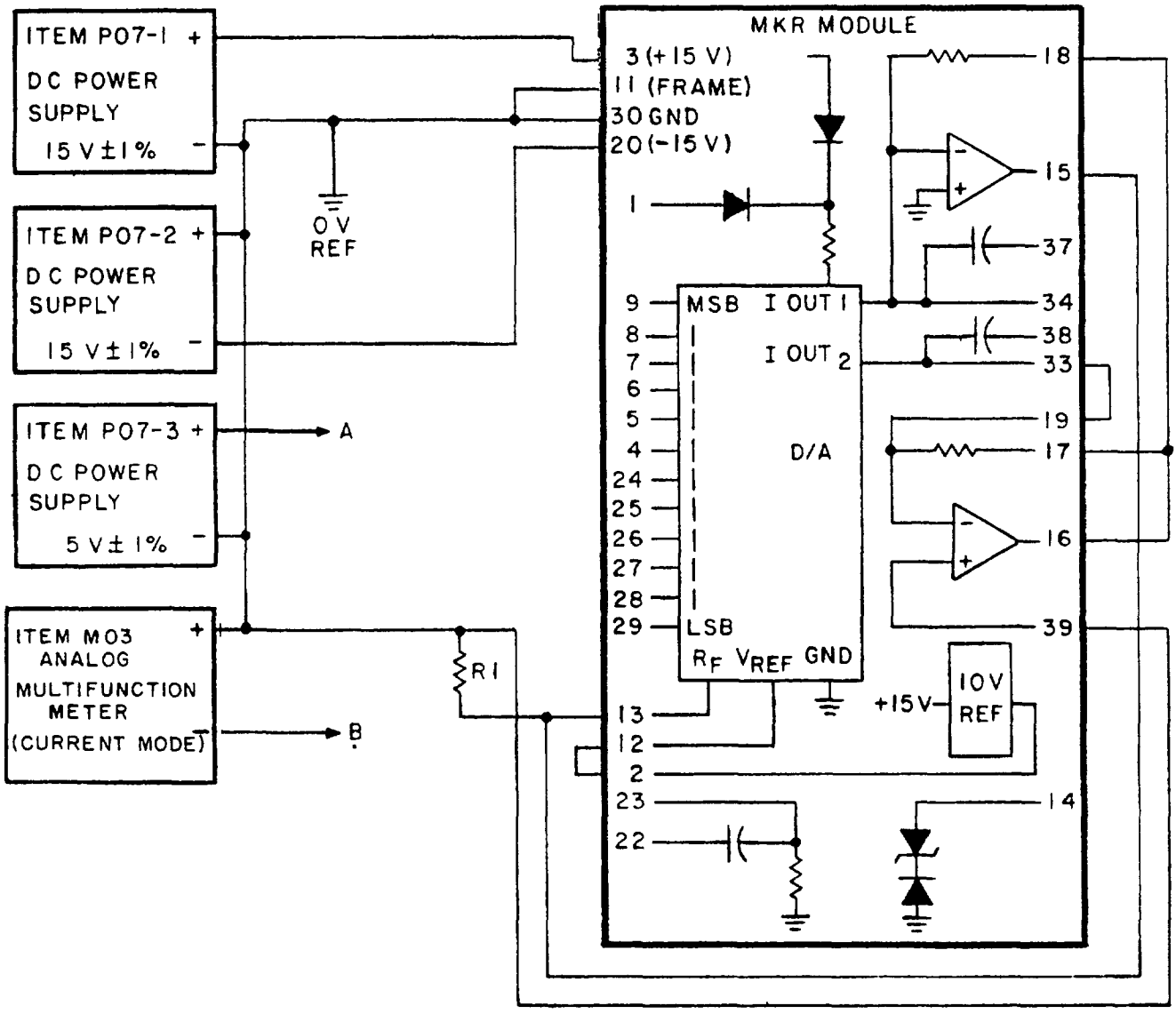
FIGURE 23 Digital input current (high-level) test circuit for table XXIII

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TABLE XXIV. Digital input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	Digital input current, low level	5,6,7,8,9,24,25,26,27, 28,29	4	-1	1	µA	24
W2		4,6,7,8,9,24,25,26,27, 28,29	5				
W3		4,5,7,8,9,24,25,26,27, 28,29	6				
W4		4,5,6,8,9,24,25,26,27, 28,29	7				
W5		4,5,6,7,9,24,25,26,27, 28,29	8				
W6		4,5,6,7,8,24,25,26,27, 28,29	9				
W7		4,5,6,7,8,9,25,26,27, 28,29	24				
W8		4,5,6,7,8,9,24,26,27, 28,29	25				
W9		4,5,6,7,8,9,24,25,27, 28,29	26				
W10		4,5,6,7,8,9,24,25,26, 28,29	27				
W11		4,5,6,7,8,9,24,25,26, 27,29	28				
W12		4,5,6,7,8,9,24,25,26, 27,28	29				

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NOTE: R1 = 10 kΩ ±1%, 1/8 W.

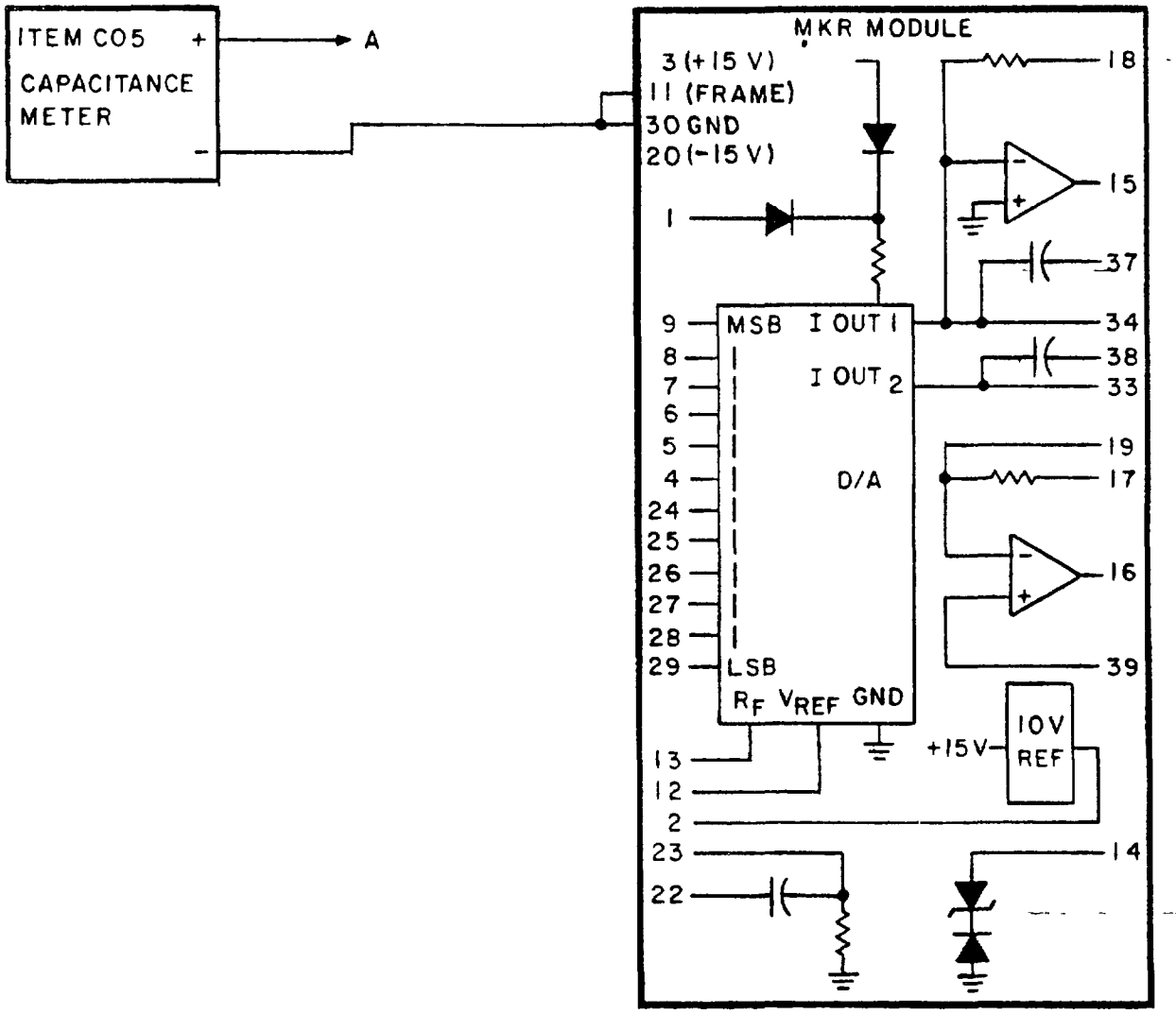
FIGURE 24. Digital input current (low-level) test circuit for table XXIV

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TABLE XXV. Digital input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
X1	Digital input capacitance	4	2	15	pF	25
X2		5				
X3		6				
X4		7				
X5		8				
X6		9				
X7		24				
X8		25				
X9		26				
X10		27				
X11		28				
X12		29				

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NOTES:
1. Null out the effect of the test leads prior to connecting the module.
2. Do not attach a test connector or wires to any of the module pins other than pins 11 and 30 and the pin under test. Stray capacitance added to other pins may affect the reading.

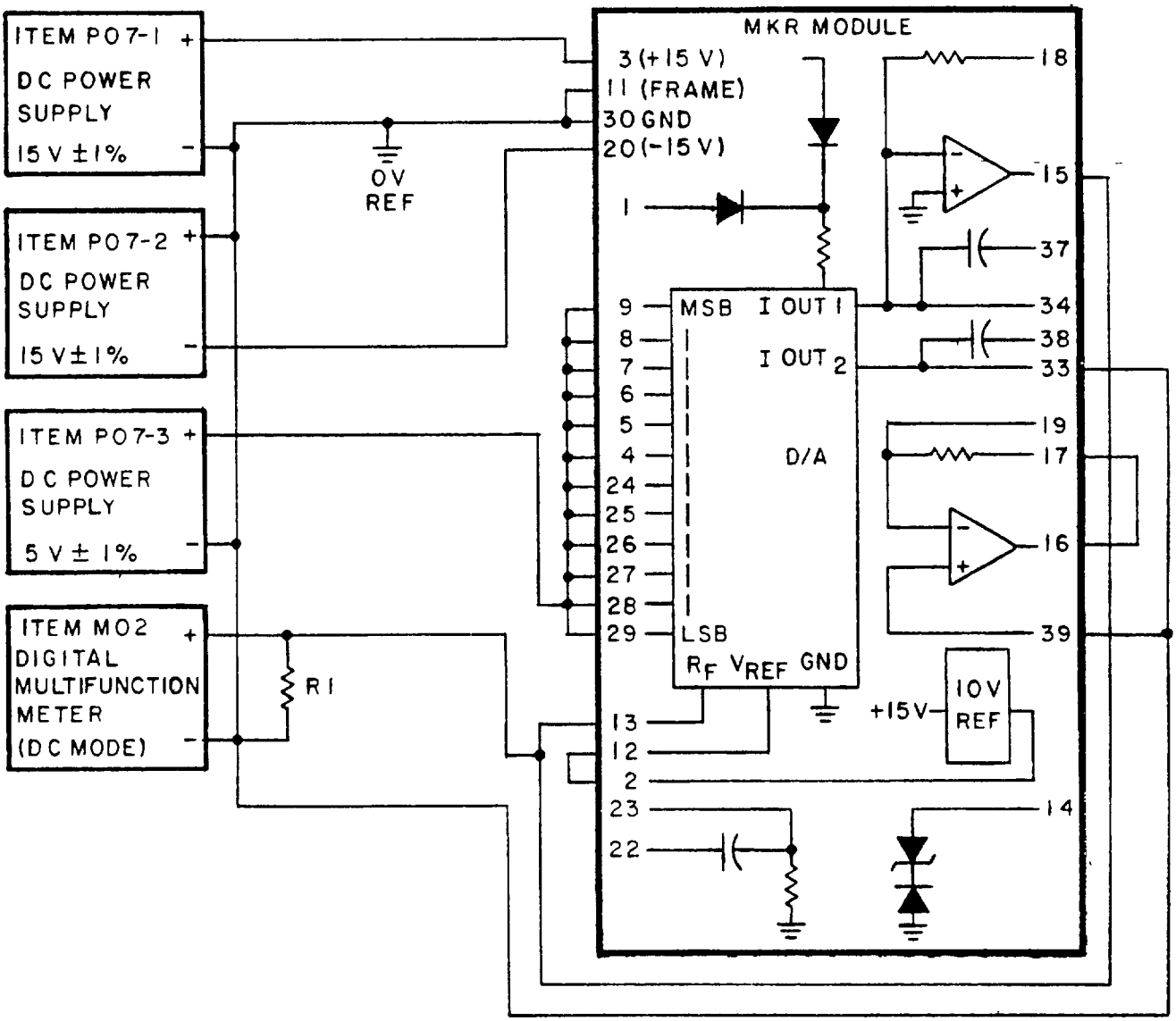
FIGURE 25. Digital input capacitance test circuit for table XXV.

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TABLE XXVI. Power supply sequence test.

Test condition	Parameter	Turn-on sequence			Initial limits 25°±5°C ambient		Unit	Test figure
		1	2	3	Min	Max		
Z1	Power supply sequence	P07-1	P07-2	P07-3	-9.8	-10.2	V	26
Z2	↓	P07-1	P07-3	P07-2	↓	↓	↓	↓
Z3		P07-2	P07-1	P07-3				
Z4		P07-2	P07-3	P07-1				
Z5		P07-3	P07-1	P07-2				
Z6		P07-3	P07-2	P07-1				

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. Turn on items PO7-1, PO7-2 and PO7-3 in the order shown in table XXVII. After each sequence verify that the output meets the limits in table XXVI.

FIGURE 26 Power supply sequence test circuit for table XXVI.

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